# Modeling electrical losses in CsPbI3 solar cells: a guardrailed prior-predictive interval at 18.0%

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***Abstract:*** *Stacked idealisations inflate SCAPS-1D CsPbI3 projections: on one fully-worked CsPbI3/CBTS stack, literature $C_r = 10^{-10}$ $cm^3/s$ alone cuts the 24.95% permissive-box ceiling to 21.86%, below the 22.02% certified record (both numbers conditional on the capped density of states). Sampling the frozen prior volume gives a prior-predictive median of* ***18.0%*** *with a 5–95% interval of* ***15.6–20.0%****—an interval under stated priors, not a confidence interval, with every measured record above its 95th percentile. Interface sensitivity is regime-dependent ($2S\tau/d$ crossing unity); a lifetime-anchored hindcast overshoots its device, bounding transfer error. All definition files, priors, sweep receipts, and raw curves are publicly archived.*

**Keywords:** CsPbI3; perovskite solar cell; SCAPS-1D; CBTS; reproducibility; device simulation

## 1.0 Introduction

All-inorganic CsPbI3 perovskite solar cells combine the thermal stability lacking in hybrid organic–inorganic perovskites with a bandgap near 1.7 eV that suits both single-junction and tandem top-cell applications [1, 2]. After years below 19%, buried-interface engineering carried laboratory efficiencies past 20% in 2024 [3] and past 22% recently: 22.02% certified (22.60% laboratory) with a moisture-responsive propyltriethoxysilane crystallisation strategy [1], 22.05% by fluorination-assisted interfacial dipole engineering [2], 21.71% with dual-interface and bulk co-passivation at an open-circuit voltage of 1.27 V [4], and 21.43% certified for inverted architectures [5]. Each record was won at the interfaces and in the bulk defect chemistry, which motivates a simulation study that treats those exact levers honestly.

The one-dimensional solar cell capacitance simulator (SCAPS-1D) is the workhorse for such pre-experimental screening, and the ITO/TiO2/CsPbI3/CBTS (copper barium thiostannate) stack of Hossain et al. [6] is the most-prescribed CsPbI3 baseline, progressing from 17.90% over 96 screened transport-layer combinations [7] to 19.06% after optimisation. Subsequent simulation studies report up to 24.17% on the same stack [8] and 24.24% on ZnO/PTAA [9], while large machine-learning-coupled screens now exceed 27% on other absorbers [10, 11]. Three problems accompany this inflation. First, nearly every study optimises one variable at a time (OVAT), so parameter interactions—notably absorber thickness versus defect-limited diffusion length—are invisible by construction. Second, interface recombination is removed in at least one case (stated explicitly in [9]) and defect densities are hard to verify without published parameters, which neither of the two 24%-class studies prints; without published parameters there is nothing to reproduce, so no cross-group adjudication is attempted. Third, scripted sweeps mix unit systems: SCAPS definition files store SI units while script 'set' commands expect practical units, so SI-valued inputs silently inject a $10^6\times$ overdose and fabricate ghost cells (scripting conventions are documented in Appendix A).

The diagnosis of this inflation now exists. A survey of more than 250 SCAPS-1D perovskite architectures finds a quarter of MAPbI3 reports exceeding the best experiments by over 20% (over nine-tenths for MASnI3), with a quarter of reported open-circuit voltages breaching the

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Shockley–Queisser bound outright; the causes are unrealistic inputs—above all the zero radiative-recombination default, unphysical defect floors and mobilities, and the isolated optimisation of inter-dependent parameters—compounded by SCAPS's optical defaults [12]. The same group has since supplied the optical-loss half of the fix, a wavelength-dependent framework toward realistic projection [13], together with a five-item standard simulation protocol: realistic inputs including the radiative coefficient, full reporting of inputs, transfer-matrix effective spectra, wavelength-dependent absorption, and an SQ check with explained deviations.

What is missing is the electrical-loss half, worked end-to-end on one stack with every assumption priced and every file archived. This paper is that companion: each protocol item maps to one row of Table 22, and each overestimation cause maps to one computed sweep.

This work contributes (i) a five-factor guardrailed grid screen (96 runs) plus refinement (36 runs, 33 converged) locating the surveyed-affinity best at 23.50%; (ii) a joint ETL/HTL-affinity map (36 runs) locating the permissive-box ceiling at **24.95%** ($\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV), with an exploratory 104-configuration search peaking 1.87 points lower among guardrail-satisfying runs; (iii) an interface dose–response reported in portable $\tau/S$ alongside raw densities; (iv) scripted series- and shunt-resistance accounting including the routine-defect case; (v) pricing of the radiative-recombination default and the mobility values; and (vi) a coupled $E_g$–$\chi$ treatment demonstrating the protocol's ban on isolated optimisation, plus a Morris screen with paired-trajectory bootstrap reported with explicit range-dependence. Run counts and receipts are itemised in Table 23; every definition file, sweep receipt, and raw curve is archived for independent reproduction.

## 2.0 Literature: a decade of CsPbI3 progress

Black-phase CsPbI3 has been stabilised stepwise since 2016: quantum-dot-induced $\alpha$-phase retention at 10.8% [14], thermodynamically stable $\gamma$-films [15], graded-gap 14.4% devices [16], and compositionally engineered 13–14% cells [17, 18], all on TiO2 with Spiro-OMeTAD, PTAA or carbon contacts [19–21]. The 2019 $\beta$-phase result at 18.4% [22] stood until buried-interface engineering carried the field past 20% (20.6%, 20.1% stabilised [3]), followed by the 2025–2026 interface-engineering wave (Figure 1): fluorinated dipoles (22.05% [2]), dual-interface plus bulk co-passivation with Voc 1.27 V (21.71% [4]), moisture-responsive crystallisation (22.60% laboratory, 22.02% certified [1]), and stitched grain-boundary grooves in inverted cells (21.43% certified [5]). The certified frontier is therefore 22.02% (Dai; laboratory best 22.60%; Qiu's 22.05% is uncertified), and every recent jump came from defect and interface control—the precise levers a simulation study must treat honestly.

Simulation studies trace a parallel inflation curve (Table 1). The CBTS screening programme reached 17.90% over 96 transport-layer combinations [7] and 19.06% after full optimisation [6], the baseline we reproduce. Later claims report higher efficiencies on differing stacks: 23.10% with ZnO/Spiro in one report [23], 24.17% on our exact stack [8], and 24.24% on ZnO/PTAA without interface layers (stated) [9]. The broader SCAPS literature shows the same pattern across absorbers—MAPbI3 at 19.30% with inorganic contacts [24], gradient-doped FAPbI3 at 20.34% [25], a 28.38% bifacial triple-cation cell [11], and an explicitly idealised 40.17% MASnI3 result that exceeds the Shockley–Queisser bound [26, 27]. Section 11.0 states the common reporting standard these numbers would need as a concrete checklist—interface status with dose level, defect floors with cross-sections—and demonstrates it on our own cell, since the published parameter tables are too incomplete to re-place anyone else's.

## 3.0 Simulation framework and guardrails

Devices were simulated in SCAPS-1D v3.3.10 under AM1.5G illumination (100 mW/cm$^2$) at 300 K unless stated. The baseline reproduces Hossain et al. [6]: ITO front contact (4.0 eV), 30 nm

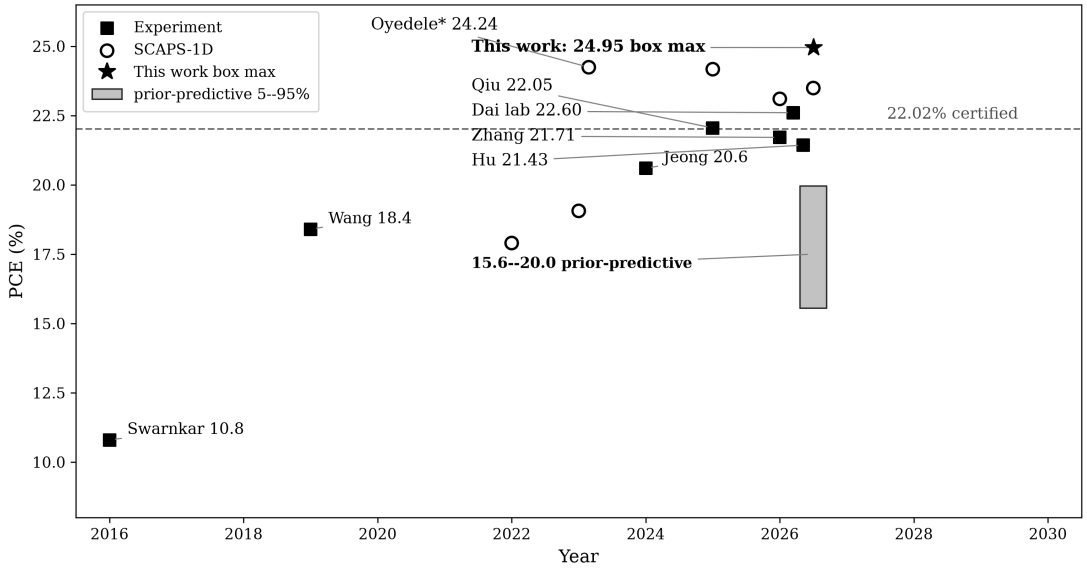



**Figure 1:** CsPbI3 record race. Asterisks mark simulation studies without interface layers; star = this work's box maximum (upper bound), band = prior-predictive 5–95% interval.

**Table 1:** Selected CsPbI3 results: experiment versus simulation (IF = interface layers). Our entry is flagged, not ranked: ⋆upper bound at the box corner (Tables 6, 4); the prior-predictive median is 18.0% (5–95%: 15.6–20.0%). [c]certified; others laboratory-reported.

| Study | Stack | Type | PCE (%) | IF |
|---|---|---|---|---|
| Swarnkar 2016 | TiO2/Spiro/MoOx | Exp. | 10.8 | – |
| Wang 2019 | TiO2/Spiro | Exp. | 18.4 | – |
| Shah et al. 2024 | TiO2/2D-Cs2PbI2Cl2 | Exp. | 20.6 (20.1 stab.) [3] | buried-passivated |
| Qiu 2025 | CsPbI3 + DFAz dipole | Exp. | 22.05 | passivated |
| Zhang 2026 | TiO2 dual-interface+bulk | Exp. | 21.71 | co-passivated |
| Dai 2026 | TiO2/PTABr/Spiro | Exp. | 22.60 lab / 22.02 cert. | passivated |
| Hu 2026 (Joule) | inverted HPDA | Exp. | 21.43 cert. | 2D-stitched |
| Hossain 2022 | TiO2/CBTS | Sim. | 17.90 | on |
| Hossain 2023 | TiO2/CBTS/Ni | Sim. | 19.06 | on |
| Ristiansyah 2026 | ZnO/Spiro | Sim. | 23.10 | on |
| Nazli 2025 | TiO2/CBTS/Ni | Sim. | 24.17 | unclear |
| Oyedele 2023 | ZnO/PTAA | Sim. | 24.24 | OFF |
| This work⋆ | TiO2/CBTS/Ni | Sim. | 24.95 | on |

TiO2 electron transport layer (donor density $9 \times 10^{16}$ cm$^{-3}$), CsPbI3 absorber (800 nm initial, $E_g = 1.694$ eV, $\chi = 3.95$ eV, $N_A = 10^{15}$ cm$^{-3}$, $N_t = 10^{15}$ cm$^{-3}$), 100 nm CBTS hole transport layer ($N_A = 10^{18}$ cm$^{-3}$), and Ni back contact (5.5 eV). Crucially, neutral interface recombination blocks ($N = 10^{10}$ cm$^{-2}$, $\sigma = 10^{-19}$ m$^2$) are present at both heterojunctions, and the absorber uses the coefficients-A-and-B absorption model ($A = 10^7$ m$^{-1}$eV$^{-1/2}$) so that bandgap sweeps act on the absorption edge rather than a frozen file spectrum. Our unmodified baseline yields 18.99% (Voc 1.122 V, Jsc 19.68 mA/cm$^2$, FF 86.00%), reproducing the published 17.9–19.06% band; against the printed 17.90% device (Voc 0.997 V, Jsc 21.07 mA/cm$^2$, FF 85.21% [7]) ours trades $\sim$1.4 mA/cm$^2$ of current for $\sim$0.12 V of voltage, as expected from added interface recombination with lower bulk defects—a single-scalar match is not claimed as validation. Optical settings: the definition file applies a flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO2, and the 80% back-transmission filter faces an opaque Ni contact, i.e. no back reflector—both choices are conservative-or-neutral for Jsc except in the blue, where the flat QE plateau is an upper envelope (Section 6.0).

Guardrails enforced throughout: bulk defect floor $N_t = 10^{12}$ cm$^{-3}$ (beyond best films; nothing lower), absorber doping ceiling $3 \times 10^{16}$ cm$^{-3}$, an interface floor of $10^{10}$ cm$^{-2}$ that is part of the stated box rather than a post-hoc filter (dose points below it are audit, not candidates), conduction-band offset held near zero by linking $\chi$ to $E_g$ ($\chi = 3.95 + 0.5(E_g - 1.694)$), and two energetic screens applied on inspection: a run with Voc above the detailed-balance value computed from the model's own Eg-sqrt absorption (1.371 V at 1.65 eV with $A = 10^7$ m$^{-1}$eV$^{-1/2}$ and 2.2 $\mu$m single-pass absorptance; the model sets $k_{rad} = 0$, priced in Section 6.2) or FF above Green's ideal-diode value at the computed Voc (90.5% at the ceiling point) would be discarded. Box constraints above are enforced per run by construction; the two energetic screens run as an automated gate over every archived receipt (`s34_gates.json`: 1024 entries checked—803 unique value-configurations, the rest intentional cross-batch replications—2 flagged, exactly the physical-DOS sensitivity runs already reported as screen-excluded above), so passing them is verified, not asserted. The $E_g/q - 0.28$ V rule is not used: at 1.65 eV it gives 1.370 V, identical to the radiative bound to 0.001 V, so it adds nothing. Series and shunt resistances are scriptable in SCAPS 3.3.10 (`set external.Rs/Rsh`; `scriptdescription3310.txt`), so this section reports computed parasitic sweeps. Sweeps ran in parallel on four workers with checkpointed receipts; of 132 grid runs, 126 converged and enter the analysis (6 missing-result-file worker crashes, IDs in Table 23; they cluster at low $N_A$, so the exclusions may be biased toward easier-to-converge doping).

SCAPS-1D solves the Poisson equation coupled to electron and hole continuity with drift-diffusion transport [28–30]:

$$-\frac{\partial}{\partial x}\left(\varepsilon(x)\frac{\partial V}{\partial x}\right) = q[p - n + N_D^+ - N_A^- + p_t - n_t], \tag{1}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G_n - R_n, \quad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G_p - R_p, \tag{2}$$

with Shockley–Read–Hall recombination through the specified bulk and interface defect levels. Optical generation uses the Eg-sqrt (Tauc) model $\alpha = A\sqrt{h\nu - E_g}$, so bandgap sweeps move the absorption edge self-consistently. Full input parameters are listed in Tables 2 and 3, with every guardrail and its justification in Table 4; transport-layer screenings that motivate the TiO2/CBTS choice are reviewed in [7, 31].

## 3.1 Protocol compliance

This study is designed as a worked implementation of the standard simulation protocol proposed in [12]; Table 22 carries the item-by-item mapping, summarised here. (i) Realistic inputs: every layer, defect, and contact parameter is printed with its source and operating-point validity in Tables 2–4, with the density-of-states cap (ten times below the source paper's printed values, §3) and the doping

**Table 2:** Layer input parameters (script units; *unmodified baseline* values as reproduced from Hossain et al. [6]: $\chi_{\mathrm{TiO2}} = 4.0$ eV, $N_{D,\mathrm{ETL}} = 9 \times 10^{16}$ cm$^{-3}$. Screening moves to $N_D = 9 \times 10^{17}$ and adopts $\chi_{\mathrm{TiO2}} = 3.8$ eV; those values appear in Table 6, not here). Mobilities in cm$^2$/Vs; $v_{th} = 10^5$ m/s all layers; contacts $S_n = S_p = 10^5$ m/s with Ni 5.5 eV / ITO 4.0 eV; front reflection filter 10%, back transmission filter 80%; radiative ($K_{rad}$) off by default and priced in Section 6.2; Auger recombination off.

| Layer | $d$ | $E_g$ | $\chi$ | $\varepsilon_r$ | Doping | $N_t$ | $\mu_n/\mu_p$ | $N_c/N_v$; defect $\sigma_n/\sigma_p$, $E_t$ |
|---|---|---|---|---|---|---|---|---|
| CBTS HTL | 100 nm | 1.9 | 3.6 | 5.4 | $N_A$ 1e18 | 1e15 | 30/30 | 2.2e18/1.8e19; 1e-20/1e-19, 1.5 eV |
| CsPbI3 | 800 nm | 1.694 | 3.95 | 6.0 | $N_A$ 1e15 | 1e15 | 25/25 | 1.1e19/8.2e19; 1e-19/1e-19, 0.6 eV |
| TiO2 ETL | 30 nm | 3.2 | 4.0 | 9.0 | $N_D$ 9e16 | 1e15 | 20/20 | 2.0e18/1.8e19; 1e-19/1e-18, 1.6 eV |

**Table 3:** Interface defect layers (both heterojunctions): neutral single level, $\sigma = 10^{-19}$ m$^2$ all sides, $E_t = 0.6$ eV.

| Interface | $N$ (cm$^{-2}$) | Sweep range (this work) |
|---|---|---|
| CBTS/CsPbI3 | 1e10 | $10^8$–$10^{12}$ (+ ETL/HTL-only, $\sigma \times 10$) |
| CsPbI3/TiO2 | 1e10 | $10^8$–$10^{12}$ (+ ETL/HTL-only, $\sigma \times 10$) |

**Table 4:** Guardrail values and what anchors them. "Assumption" means a modelling choice with no direct measurement behind it.

| Guardrail | Value | Anchor |
|---|---|---|
| Bulk defect floor | $\tau = 100$ $\mu$s ($N_t = 10^{12}$ cm$^{-3}$, $\sigma = 10^{-19}$ m$^2$) | lifetime guardrail, beyond best films; routine $\tau = 0.1$ $\mu$s |
| Interface density | $N_{if} = 10^{10}$ cm$^{-2}$ | assumption; $S \approx 100$ cm/s (demanding passivation literature) |
| Capture cross-section | $\sigma = 10^{-19}$ m$^2$ | assumption (SCAPS-typical; swept $\times 0.1$–$\times 10$) |
| Trap level | $E_t = 0.6$ eV mid-gap | assumption (0.3–0.9 eV sweep: $\leq 0.012$ pts) |
| $V_{oc}$ ceiling | 1.371 V at $E_g = 1.65$ eV (1.418 V at 1.70) | detailed-balance, model absorption ($k_{rad} = 0$) |
| Radiative coefficient | $k_{rad} = 0$ (priced: $-3.09/-5.12$ pts at $C_r = 10^{-10}/10^{-9}$; $-0.14/-1.02$ routine) | assumption, priced Sec. 6.2; Eq. 5 $\approx$ omission |
| Mobilities | 25/25 absorber (priced: $\leq 0.01$ ceiling, $-0.32$ routine at 10/5) | assumption, priced Sec. 6.2 |
| FF ceiling | Green's ideal at computed $V_{oc}$ | analytic bound |
| $R_s$, $R_{sh}$ basis | scripted sweeps, reported | computed, not assumed zero |

**Table 5:** The five simulation bases used in this paper. Affinities are ($\chi_{\mathrm{CBTS}}$, $\chi_{\mathrm{TiO2}}$); other parameters per Table 19 unless noted.

| Base | Definition | Carries |
|---|---|---|
| Surveyed | r7 def as reproduced: $\chi$ (3.6, 4.0), $E_g$ 1.694, 0.8 $\mu$m, $N_A/N_t = 10^{15}/10^{15}$ | baseline 18.99% |
| Prior adopted | surveyed + $N_D = 9\times10^{17}$, $E_g$ = 1.65 + $\chi$-link | dose/audit anchor 23.84% |
| Ceiling | adopted + $\chi$ (3.9, 3.7), 2.2 $\mu$m, $N_A = 3\times10^{16}$, $N_t = 10^{12}$ | 24.95% ceiling |
| Routine | surveyed $\chi$, 1.0 $\mu$m, $N_t = 10^{15}$ (Table 19) | reference 20.82% |
| New base (s16/17) | joint $\chi$, $E_g$ steps, $R_s$ variants | T/Eg/Rs anchors |
| Calibration | surveyed $\chi$, $E_g$ = 1.70 + link, ITO 4.2, $N_{if} = 3\times10^{10}$ | ladder 19.3–20.7% |

ceiling stated as the optimistic modeling choice it is (measured densities sit below $10^{12}$ cm$^{-3}$; §9) rather than buried. (ii) Full reporting: bulk and interface defects are reported as *N with* $\sigma$ alongside the portable $\tau$/*S* equivalents (Table 16), and the radiative coefficient is reported as a swept assumption, not a default (Section 6.2). (iii) Effective spectra and absorption: we retain the Eg-sqrt model with the edge-moving coefficient so gap sweeps act on absorption rather than a frozen spectrum, and we carry the optical uncertainty instead of hiding it—a flat 10% front reflection with no dispersive parasitics (an upper envelope in the blue, $\sim$1 mA/cm$^2$ at stake) plus the 1.19 mA/cm$^2$ QE-integral-vs-IV gap resolved in favour of the IV number (Section 6.0); the wavelength-dependent transfer-matrix half is cited, not re-derived [13]. (iv) SQ check: the model-derived 1.371 V radiative bound is evaluated against every run post-hoc (Section 7.0), and the no-interface band's lack of margin is reported as disqualification, not rank order. (v) Explained deviations: Table 21 attributes the residual excess over measured devices to named causes (fill factor, gap) with the un-isolated systematics stated. Where the protocol asks for an explanation and we have only a bound, we say so.

Five named bases recur throughout; Table 5 defines each once, with the headline number each carries, so that scattered values always resolve to one parameter set.

## 4.0 Grid screen and permissive-box ceiling

Round A explored the full cross-product of absorber thickness (1.0–2.0 $\mu$m), acceptor density ($10^{15}$–$10^{16}$ cm$^{-3}$), bulk defect density ($10^{12}$–$10^{14}$ cm$^{-3}$), ETL donor density ($9\times10^{16}$–$9\times10^{17}$ cm$^{-3}$), and bandgap (1.694/1.65 eV with linked affinity)—96 runs (93 converged, 3 worker crashes) in a single parallel batch, rather than sequential OVAT slices. The winner (2.0 $\mu$m, $N_A = 10^{16}$ cm$^{-3}$, $N_t = 10^{12}$ cm$^{-3}$, $N_{D,ETL} = 9\times10^{17}$ cm$^{-3}$, $E_g = 1.65$ eV) reached 23.47%. Full parameters are collected in Table 6. Round B refined thickness (1.8–2.2 $\mu$m), gap (1.65–1.72 eV), and doping around this point (36 runs, 33 converged) and reached 23.50% (2.2 $\mu$m, $N_A = 3\times10^{16}$ cm$^{-3}$, $N_t = 10^{12}$ cm$^{-3}$, $E_g = 1.65$ eV, $\chi = 3.928$ eV). A subsequent ETL-affinity sweep showed the surveyed 4.0 eV is not optimal: 3.9 eV gives 23.80%, 3.8 eV gives 23.82–23.84% across two batches (23.8241 in `s7_r3` and `s11`, 23.8365 in `s8`; the 0.012 spread is inside the 0.017-point IV-step uncertainty above, and we quote 23.84% for the batch that produced the gap-step and $E_t$ analysis, against the surveyed anchor 23.5041), while 4.1 and 4.2 eV collapse to 21.92% and 19.87%. Because the two affinities interact through the interface activation energy, we then mapped them jointly at guardrailed interfaces (36 runs; Table 7 prints every second $\chi_{\mathrm{CBTS}}$ row plus the 3.95–4.10 eV extension): at fixed $\chi_{\mathrm{TiO2}}$, efficiency rises monotonically with $\chi_{\mathrm{CBTS}}$ through 3.90 eV, and the peak extension (3.95–4.10 eV) turns over past $\sim$3.95 eV, so the optimum is interior, not edge-sitting. A dedicated extension toward lower ETL affinities confirms the interior claim on that side too: at $\chi_{\mathrm{TiO2}} = 3.6$ the map gives

**Table 6:** Ceiling-point device parameters (script units) and result: highest evaluated of 180 guardrail-satisfying configurations on named geometries, not a proven global optimum.

| Layer | $d$ | $E_g$ (eV) | $\chi$ (eV) | Doping (cm$^{-3}$) | $N_t$ (cm$^{-3}$) |
|---|---|---|---|---|---|
| CBTS HTL | 100 nm | 1.9 | 3.9 | NA 1e18 | 1e15 |
| CsPbI3 absorber | 2.2 $\mu$m | 1.65 | 3.928 | NA 3e16 | 1e12 |
| TiO2 ETL | 30 nm | 3.2 | 3.7 (ceiling adoption; surveyed 4.0) | ND 9e17 | 1e15 |
| Interfaces (both) | – | – | – | – | 1e10 cm$^{-2}$ |
| Result | Voc 1.321 V, Jsc 21.20 mA/cm$^2$, FF 89.11%, PCE 24.95% | | | | |

24.62/24.85/24.91% ($\chi_{\mathrm{CBTS}} = 3.7/3.8/3.9$), falling to 23.55/23.76/23.84% at 3.5 eV, so the ridge peaks between 3.6 and 3.7 eV and descends on both sides (receipts `s30_crchi.jsonl`, extension columns). The permissive-box ceiling is **24.95%** at $\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV (Voc 1.321 V, Jsc 21.20 mA/cm$^2$, FF 89.11%; Figure 2), verified digit-identical with explicitly set $N_{if} = 10^{10}$ cm$^{-2}$ (24.9464 both ways, receipt `s22_jointIF.json`) and falling to 24.24% at $10^{11}$ cm$^{-2}$ (24.2377, same receipt), confirming the interfaces are active at the optimum. An exploratory 104-configuration search (LHS plus pattern search over thickness, $\log N_A$, $\log N_t$, $\log N_D$, $E_g$, $\log N_{if}$, $\chi_{\mathrm{TiO2}}$) was run over a deliberately wider box than the guardrailed one—$\log N_{if}$ down to $10^8$ and $\chi_{\mathrm{TiO2}}$ from 3.8 eV—so that it cannot bracket the ceiling and is reported only as an exploration: 95 of 104 runs converged, 69 sit below the $10^{10}$ cm$^{-2}$ interface floor and none samples $\chi_{\mathrm{TiO2}} < 3.8$ eV. Its overall peak is 23.76% (itself a sub-guardrail cell at $N_{if} = 4.6 \times 10^9$ cm$^{-2}$), and 23.08% among its 25 runs satisfying both the interface floor and the stated $N_A \leq 3 \times 10^{16}$ cm$^{-3}$ ceiling (26 meet the interface floor alone; the sampler overshot the doping ceiling by 2% in one run)—below every grid-screened best, but not a test of it. 24.95% therefore stands as the highest of the 180 guardrail-satisfying configurations we evaluated on named geometries—Round-A 96 (`roundA.jsonl`), Round-B 36 (`roundB.jsonl`), the 36 joint-affinity map cells of `s11_joint_chi.json` (excluding its 16 work-function variants, which probe contacts), and the 12 affinity-adoption runs of `s8_champ38.json`; 174 converged—plus routine-defect and audit batches that probe rather than optimise—not as a proven global optimum. The full batch inventory, including the ITO/Ni variants, the s16/s17 extensions, and every retry, is itemised in Table 23. Two features of this point matter. First, it sits at the corner of the allowed box in every dimension except the affinities, which are themselves aggressive ($\chi_{\mathrm{TiO2}} = 3.7$ eV below the typical 4.0–4.2 eV range; $\chi_{\mathrm{CBTS}} = 3.9$ eV above the surveyed 3.6 eV): this is a permissive-edge upper bound, and Section 9.0 prices each edge. Second, the highest grid point at the literature gap $E_g = 1.694$ eV reaches 22.506%, coincidentally equal at two decimals to the 22.514% of the $E_g = 1.70$ eV attainable-gap run on a different geometry (they differ by 0.008 points, inside step uncertainty), so roughly one point comes from the gap choice alone. Every top-ten configuration carries $E_g = 1.65$ eV, gaining $\sim$1.5 mA/cm$^2$ over 1.694 eV; the ceiling cell presents a +0.23 eV ETL-side electron spike that raises the interface recombination activation energy (Section 5.0).

From diffusion estimates ($\mu_n = 25$ cm$^2$/Vs, $\sigma = 10^{-19}$ m$^2$): $L \approx 80$ $\mu$m at $10^{12}$ cm$^{-3}$, $\approx$8 $\mu$m at $10^{14}$ cm$^{-3}$, $\approx$2.5 $\mu$m at $10^{15}$ cm$^{-3}$. The map (Figure 3) is therefore nearly thickness-insensitive at $10^{14}$ cm$^{-3}$ (near-vertical contours)—a thickness collapse there cannot be diffusion-limited, so we do not claim one. The thickness optimum moves with defect level, which is why the single-variable sweep had to be repeated. A two-factor thickness $\times$ $N_t$ factorial at the ceiling base gives, at $N_t = 10^{15}$ cm$^{-3}$, 20.11% (0.5 $\mu$m), 20.76% (0.8 $\mu$m), 20.86% (1.0 $\mu$m) and 20.81% (1.5 $\mu$m), against 21.67/22.42/22.53/22.57% at $10^{14}$ cm$^{-3}$: the same 0.7–0.9 points of thickness structure at both defect levels, with the peak moving from 1.5 $\mu$m at the cleaner film to 1.0 $\mu$m at routine defects—the interaction is real but small, and we claim only that the optimum shifts with $N_t$, not that thickness dominates. ETL donor density confirms the literature optimum ($9 \times 10^{17}$ cm$^{-3}$) and transport-layer thicknesses are insensitive (100/30 nm retained), both consistent with [6]. The current–voltage curve (Figure 2) is square and artefact-free, with a Voc deficit of 0.33 V against $E_g/q$ at the ceiling (0.40 V at the prior $\chi_{\mathrm{TiO2}} = 3.8$ eV base).

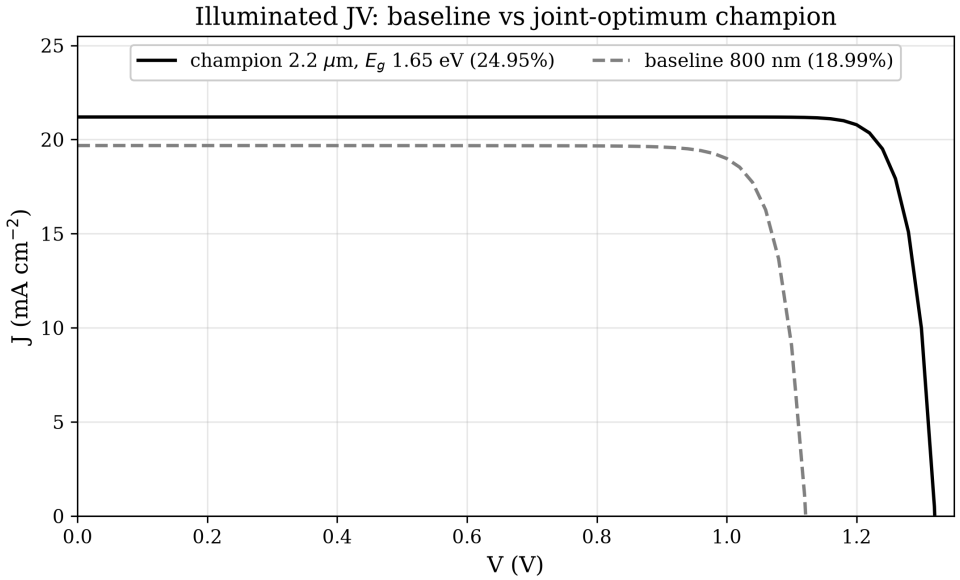



**Figure 2:** Illuminated current–voltage curves: baseline (800 nm, 18.99%) versus permissive-box ceiling (2.2 $\mu$m, 24.95%).

**Table 7:** Joint affinity map at guardrailed interfaces (PCE %; ceiling geometry; peak extension 3.95–4.10 eV turns over past ∼3.95 eV).

| $\chi_{\mathrm{CBTS}} \setminus \chi_{\mathrm{TiO2}}$ | 3.7 | 3.8 | 3.9 | 4.0 |
|---|---|---|---|---|
| 3.50 | 22.10 | 22.10 | 22.10 | 22.07 |
| 3.60 | 23.83 | 23.82 | 23.80 | 23.50 |
| 3.70 | 24.67 | 24.64 | 24.53 | 23.77 |
| 3.80 | 24.89 | 24.86 | 24.67 | 23.81 |
| 3.90 | **24.95** | 24.91 | 24.74 | 23.81 |
| 3.95 | 24.94 | 24.90 | – | – |
| 4.00 | 24.81 | 24.79 | – | – |
| 4.10 | 23.25 | 23.25 | – | – |

The 5.96-point stacked gain decomposes along the screening path (Table 6 parameters in sequence, illustrative ordering only): thickness 800 nm to 1.5 $\mu$m adds 0.19 points (19.18%), acceptor doping to $10^{16}$ cm$^{-3}$ adds 0.93 points (20.11%), bulk-defect reduction to $10^{12}$ cm$^{-3}$ adds ∼1.6 points (∼21.7%), the joint $E_g$–$\chi$ move to 1.65 eV with ETL donor optimisation contributes ∼1.8 points (to the 23.50% surveyed best), and the joint affinity adoption adds the final 1.45 points (modulo rounding). This sequential attribution is order-dependent and illustrative, not causal. Open-circuit voltage climbs 1.122 to 1.321 V while the fill factor rises 86.0 to 89.1%, the signature of SRH-lifetime recovery plus interface-activation recovery rather than band-shift artefacts.

Round A's top ten (Table 8) all carry $E_g = 1.65$ eV; the top seven sit at $N_t = 10^{12}$ cm$^{-3}$ while rows 8–10 at $N_t = 10^{13}$ cm$^{-3}$ trail by ∼0.4 points through Voc loss (1.233 to 1.221 V). Round B refined thickness (1.8–2.2 $\mu$m), gap (1.65–1.72 eV) and doping (36 runs) and moved the best from 23.47% to 23.50% (coarse grid), i.e. the landscape is converged rather than climbing.

Round B's refinement is documented in full because its plateau is itself evidence: the top ten span only 23.32–23.50% (Table 9), with thickness 1.8–2.2 $\mu$m and $N_A$ (1–3)$\times 10^{16}$ cm$^{-3}$ numerically indistinguishable at grid resolution once $N_t = 10^{12}$ cm$^{-3}$ and $E_g = 1.65$ eV are fixed (the ten span 23.32–23.50%, a 0.19-point range against a 0.017-point coarse/fine IV-step uncertainty, so the leading cluster cannot be separated even though the 10th entry trails observably). The $N_t = 3 \times 10^{12}$ cm$^{-3}$

**Table 8:** Round A leaderboard (top ten of 96; $N_{D,\mathrm{ETL}} = 9 \times 10^{17}$ cm$^{-3}$; $\chi_{\mathrm{TiO2}} = 4.0$ eV surveyed throughout).

| $d$ ($\mu$m) | $N_A$ | $N_t$ | $E_g$ | Voc | Jsc | PCE |
|---|---|---|---|---|---|---|
| 2.0 | 1e16 | 1e12 | 1.65 | 1.233 | 21.18 | 23.47 |
| 1.8 | 1e16 | 1e12 | 1.65 | 1.233 | 21.16 | 23.45 |
| 1.5 | 1e16 | 1e12 | 1.65 | 1.234 | 21.11 | 23.40 |
| 1.5 | 1e15 | 1e12 | 1.65 | 1.233 | 21.11 | 23.29 |
| 2.0 | 1e15 | 1e12 | 1.65 | 1.233 | 21.18 | 23.29 |
| 1.0 | 1e16 | 1e12 | 1.65 | 1.234 | 20.90 | 23.18 |
| 1.0 | 1e15 | 1e12 | 1.65 | 1.233 | 20.90 | 23.11 |
| 1.8 | 1e16 | 1e13 | 1.65 | 1.221 | 21.16 | 23.04 |
| 2.0 | 1e16 | 1e13 | 1.65 | 1.219 | 21.18 | 23.03 |
| 1.5 | 1e16 | 1e13 | 1.65 | 1.223 | 21.11 | 23.03 |

**Table 9:** Round B leaderboard (top ten of 36; $N_{D,\mathrm{ETL}} = 9 \times 10^{17}$ cm$^{-3}$).

| $d$ ($\mu$m) | $N_A$ | $N_t$ | $E_g$ | PCE |
|---|---|---|---|---|
| 2.2 | 3e16 | 1e12 | 1.65 | 23.50 |
| 2.0 | 3e16 | 1e12 | 1.65 | 23.49 |
| 2.2 | 1e16 | 1e12 | 1.65 | 23.48 |
| 2.0 | 1e16 | 1e12 | 1.65 | 23.47 |
| 1.8 | 3e16 | 1e12 | 1.65 | 23.47 |
| 2.2 | 3e16 | 3e12 | 1.65 | 23.45 |
| 1.8 | 1e16 | 1e12 | 1.65 | 23.45 |
| 2.0 | 3e16 | 3e12 | 1.65 | 23.44 |
| 1.8 | 3e16 | 3e12 | 1.65 | 23.42 |
| 2.2 | 1e16 | 3e12 | 1.65 | 23.32 |

entries trail the matched $10^{12}$ cells by 0.05–0.16 points (0.05 at the ceiling corner, 0.16 on the $N_A = 10^{16}$ branch), confirming the defect floor binds the optimum.

## 5.0 Band alignment

Figure 4 constructs the equilibrium alignment from the ceiling-point parameters. The absorber conduction edge ($-3.928$ eV) lies 0.23 eV *below* the TiO2 edge ($-3.70$ eV), so electrons face a spike entering the ETL. A correction on the hole side first: the local conduction offset at an interface is fixed by $\Delta\chi$ (0.028 eV here) at any bias—the $\sim$0.38 eV step read off the in-device profile is built-in band bending from the p/p$^+$ doping step (CBTS $N_A = 10^{18}$ against absorber $3 \times 10^{16}$ cm$^{-3}$), back-surface-field-like, not an offset opening. It still blocks electrons, but it exists because of the doping, which ties the $\chi_{\mathrm{CBTS}} = 3.9$ optimum to the $3 \times 10^{16}$ cm$^{-3}$ doping assumption (four orders above measured densities; the NA$\times\chi_{\mathrm{CBTS}}$ interaction is priced in Section 9.0). The data support reading this as recombination physics, not transport: the interface recombination activation energy $E_a = E_{c,\mathrm{abs}} - E_{v,\mathrm{HTL}}$ grows from 1.47 eV at $\chi_{\mathrm{CBTS}} = 3.5$ to 1.87 eV at 3.9 eV, and efficiency rises monotonically through zero hole-side offset (22.10% to 24.95% at $\chi_{\mathrm{TiO2}} = 3.7$; Table 7)—a transport reading predicts the barrier end should be worst, and it is the best. On the hole side, the absorber valence edge ($-5.578$ eV) lies 0.22 eV *above* the CBTS edge ($-5.80$ eV) at the ceiling, past the ideal $-0.3$ to $-0.1$ eV window on the barrier side—yet efficiency peaks there, which a transport reading cannot produce. The same activation logic applies to the ETL side, which is why the spike buys rather than costs efficiency. Both optimum affinities sit outside the surveyed values ($\chi_{\mathrm{TiO2}} = 3.7$ below the typical 4.0–4.2 eV; $\chi_{\mathrm{CBTS}} = 3.9$ above the surveyed 3.6 eV) and are reported as ceiling

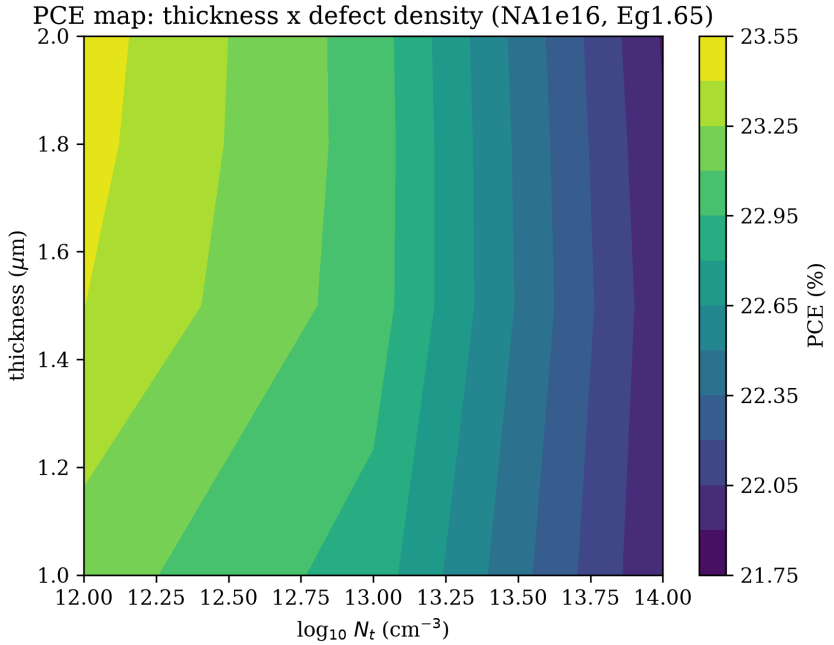



**Figure 3:** Efficiency map over absorber thickness and bulk defect density (slice at $N_A = 10^{16}$ cm$^{-3}$, $E_g = 1.65$ eV).

adoptions priced in Section 9.0, not surveyed values.

The electron-blocking question is answered in-device, not from vacuum levels. Band diagrams saved under bias at the ceiling point (receipts `JEB_v115`, `JEB_v130`; quasi-Fermi splitting 1.151/1.300 V confirming the operating points) show the vacuum-input 0.03 eV CBTS/absorber conduction offset opening to ∼0.38 eV in-device at short circuit (CBTS $E_c = 1.825$ vs absorber 1.445 in solver coordinates), so the HTL does block electrons: near maximum power (1.15 V) the electron current at the back contact is 0.01 against a hole current of 21 (same solver units), i.e. three orders of leakage suppression. The voltage is nevertheless lost at the back: at 1.30 V the back-region recombination peak reaches $1.6 \times 10^{20}$ against $4.4 \times 10^{18}$ generation, with electron leakage rising to 3.3 against 13 holes. The ceiling cell's voltage deficit is therefore a back-interface recombination budget item, priced by the dose–response of Section 7.0, not a transport barrier—and the short-circuit profiles of Section 6.0, which show none of this, cannot constrain it. Whether the $\chi_{\mathrm{CBTS}} = 3.9$ optimum is partly a contact-regime artifact was tested directly: deepening the Ni work function to 5.8 and 6.0 eV at the ceiling base (archived definition variants, receipts `s26_diag.json`) moves efficiency from 24.9464% to 24.9470% and 24.9480%—unchanged against the 0.017-point resolution floor. Removing the ∼0.1–0.3 eV hole-extraction barrier buys nothing, so the contact does not gate the optimum and the affinity ranking stands as interface activation physics, not contact artifact. Nor does the $\chi_{\mathrm{CBTS}}$ optimum depend on the optimistic doping it was found under: at $N_A = 10^{15}$ cm$^{-3}$ the 3.6-to-3.9 step still gains 0.77 points (23.51 to 24.28%) at the same 3.9 location, against 1.12 points at $3 \times 10^{16}$ (23.83 to 24.95%; receipts `s33_nachi.json`). The optimum location transfers to realistic doping; only its height does not.

## 6.0 Device characterisation and assumption pricing

The quantum efficiency (Figure 5) is flat at ∼89–90% from 300 nm with the absorption edge redshifted from 731 nm (baseline, 1.694 eV) to 750 nm (ceiling cell, 1.65 eV), directly explaining the Jsc gain. The UV plateau assumes the definition file's flat 10% front reflection with no dispersive parasitic absorption in ITO or TiO2, so it is an upper envelope in the blue rather than a prediction;

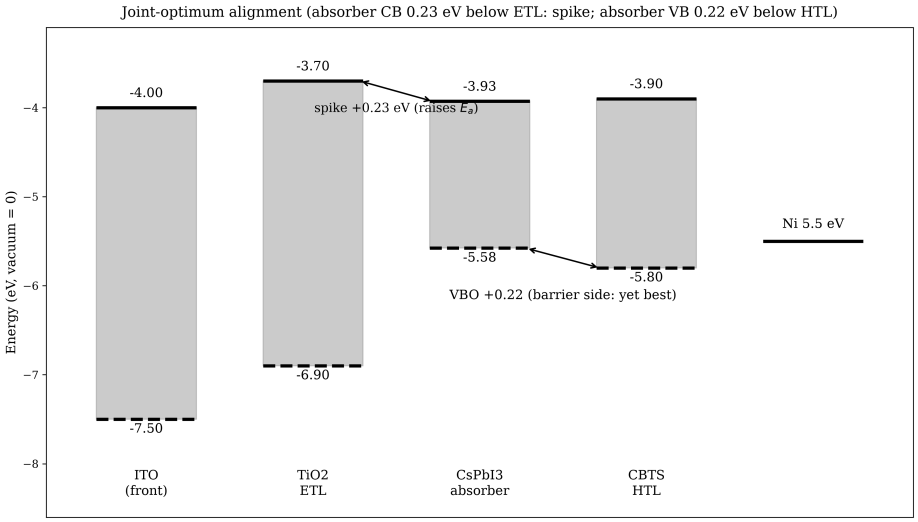



**Figure 4:** Equilibrium band alignment of the ceiling stack from input parameters (energies in eV vs vacuum).

with TiO2 absorbing above 3.2 eV, of order 1 mA/cm$^2$ of the reported Jsc is at stake in the UV, which is large next to the points debated here and is stated as an uncertainty, not corrected. Integrating the computed QE against the AM1.5G spectrum gives 22.39 mA/cm$^2$ against the 21.20 mA/cm$^2$ IV value. That 1.19 mA/cm$^2$ gap is a same-harness diagnostic, and it survives the obvious explanations: the IV Jsc reproduces to seven decimals across IV step sizes (0.02 and 0.05 V); the QE curve is byte-identical under explicit constant-photon-flux and constant-photon-power illumination modes (receipt `s27`, recovered post-hoc); and the IV run's own generation-minus-recombination integral gives 21.21 mA/cm$^2$, exactly the IV value. Two sharper tests are blocked by the harness, and reported as blocked rather than passed: loading alternate spectra (NREL AM1.5G, AM0, white 400–800, absolute or bare paths) returns the default-spectrum IV digit-identically every time, so the `load spectrumfile` key is inert in batch scripts and no spectrum discrimination was achieved; the spectrum-cutoff keys behave the same way. So the IV path is converged and internally self-consistent, while the QE sweep path over-delivers flux by $\sim$6% through a normalisation this study could not close from the script interface. The gap is therefore carried as a two-sided $\pm$1.2 mA/cm$^2$ absolute-scale systematic ($\sim$1 point on PCE): it shifts every absolute headline together and cancels in every point-difference quoted, which are all IV-path-internal. A bounding argument favours the IV side independently of any flux theory: the ideal step-absorption SQ current at 1.65 eV is 24.4 mA/cm$^2$, so with the stated 10% front reflection the ceiling is 21.96—already below the 22.39 QE integral, which therefore cannot be the collected current of this optical stack. The headlined IV value remains the lower, converged number. Capacitance–voltage analysis at 1 MHz (Figure 6, joint-affinity base) shows rising capacitance with forward bias (31 to 39 nF/cm$^2$ over 0–0.5 V); the low-bias $1/C^2$ slope recovers the input doping ($N_A = 2.96\times10^{16}$ cm$^{-3}$, $V_{bi} = 1.33$ V), a solver self-consistency check, not an independent validation. The SCAPS-extracted width (0.29 to 0.22 $\mu$m) sits an order of magnitude below the 2.2 $\mu$m absorber throughout; the parallel-plate value at 0 V is 0.17 $\mu$m (31 nF/cm$^2$, $\varepsilon_r = 6$), so the check confirms the sign and linearity of the $1/C^2$ branch, not the absorber thickness, and no depletion-width claim is made beyond it. The illuminated band diagram at short circuit (Figure 7, joint affinities) exhibits clean quasi-Fermi splitting, and generation exceeds recombination by at least two orders of magnitude at every absorber mesh point of the full 2.33 $\mu$m short-circuit profile (minimum ratio $\sim3\times10^2$; Figure 8)—but *not* at the CBTS interface plane itself, where 25 mesh points show recombination locally exceeding generation (ratio down to $\sim$0.5): the priced interface channel, visible in the profile rather than hidden by it. All profile statements in this section are at short circuit; the biased-regime budget was established in Section 5.0.

Temperature dependence (Figure 9, Table 10) follows the established trend [6]: efficiency falls

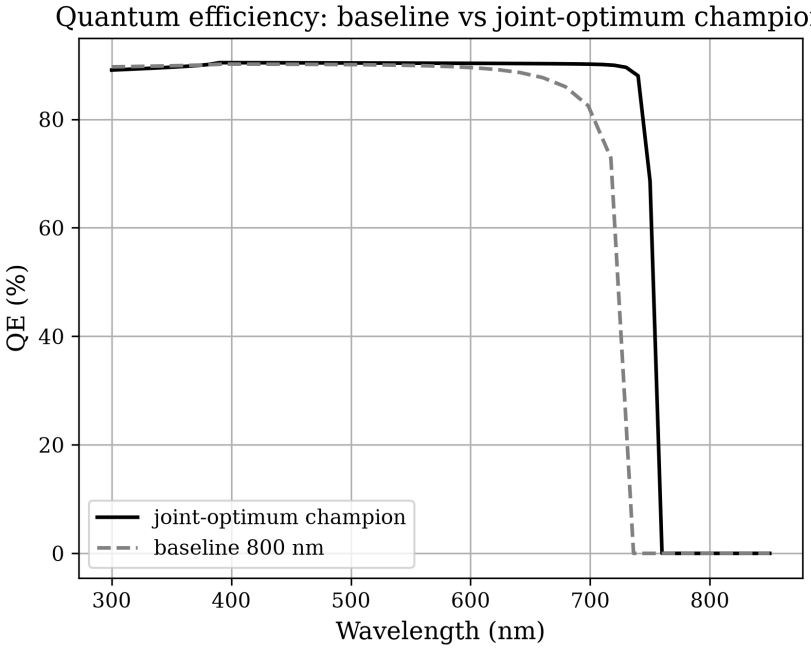



**Figure 5:** Quantum efficiency of baseline versus ceiling cell (joint-affinity base); the edge shifts 731 to 750 nm.

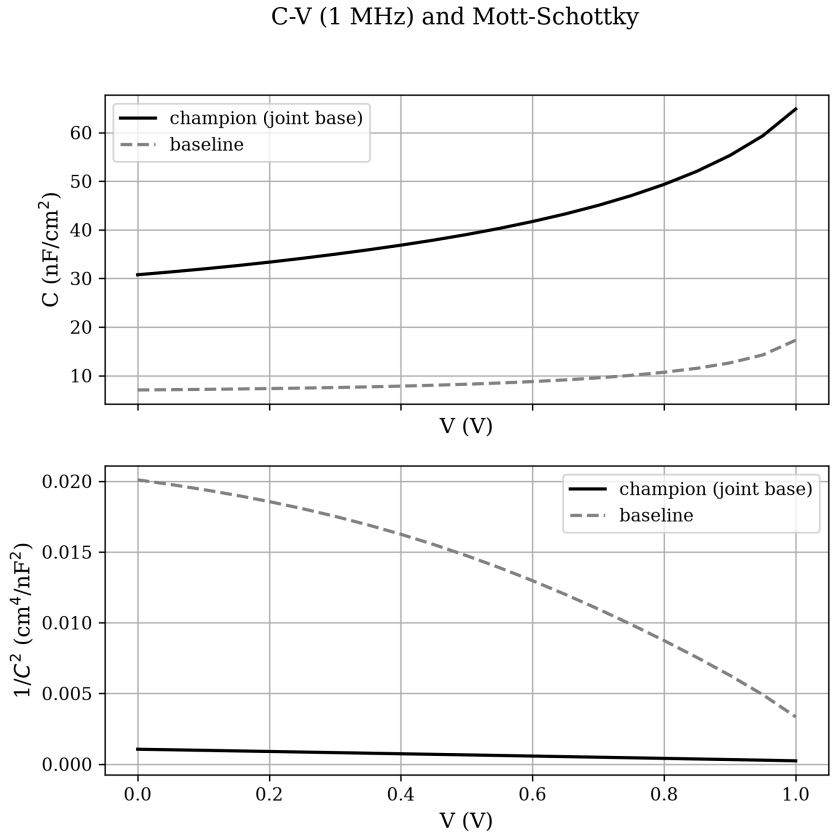



**Figure 6:** Capacitance–voltage (1 MHz) and Mott–Schottky plots for baseline and ceiling cell.

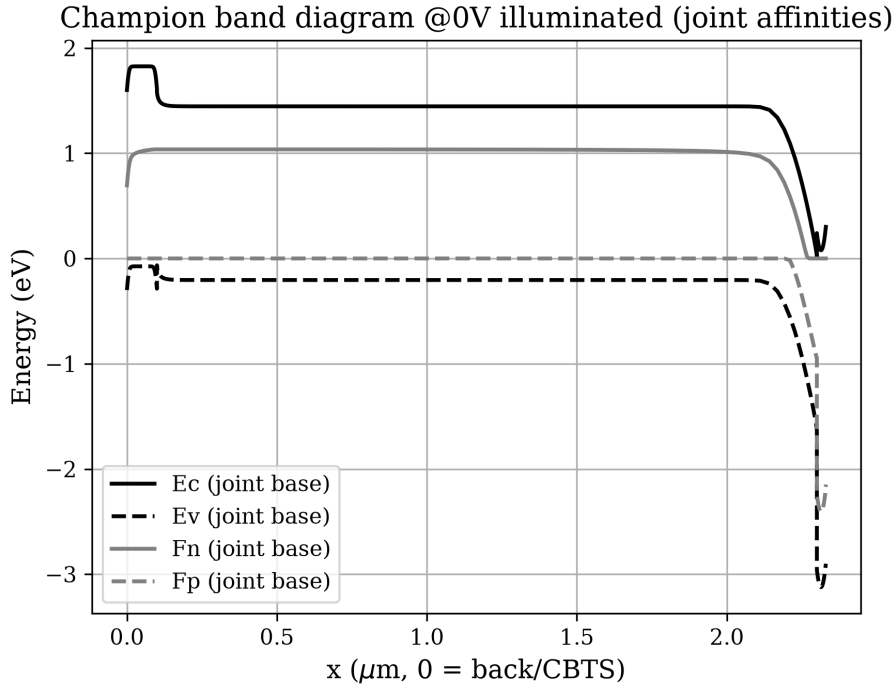



**Figure 7:** Illuminated band diagram of the ceiling cell at short circuit (x=0 at the back contact).

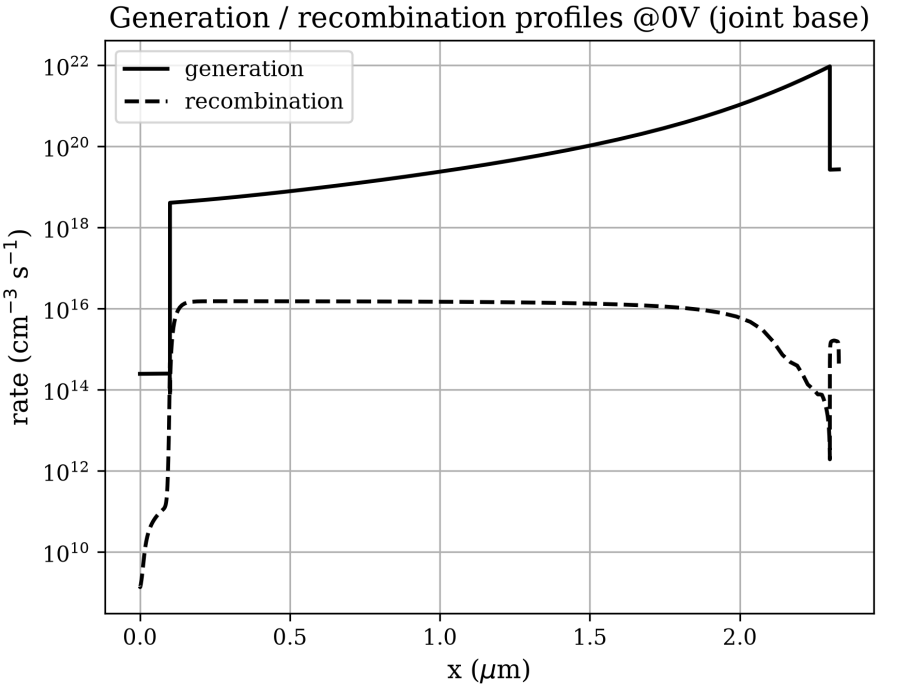



**Figure 8:** Generation and recombination profiles at short circuit (log scale).

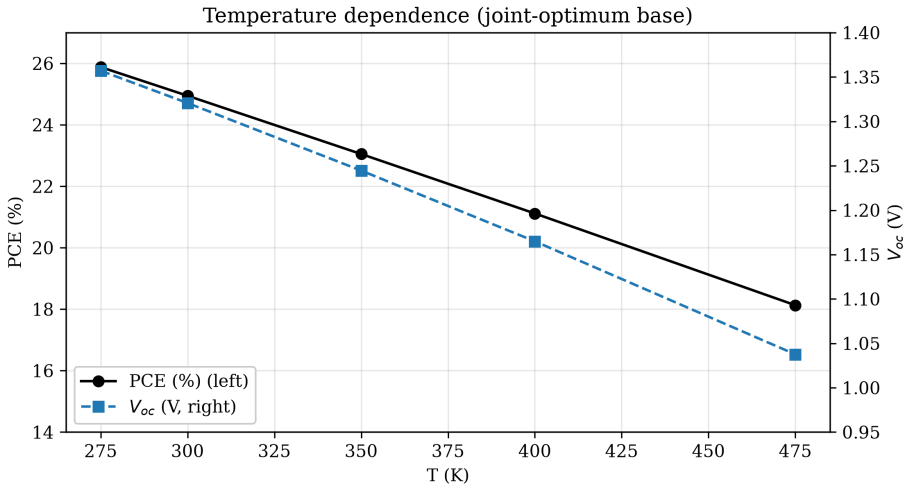



**Figure 9:** Efficiency and open-circuit voltage versus operating temperature at the ceiling base ($R_s = 0$); the 475 K end is a solver stress test that ignores $E_g(T)$ and black-phase stability.

**Table 10:** Temperature dependence at the ceiling base (1 sun; T300 = 24.9464).

| T (K) | Voc (V) | Jsc (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| 275 | 1.357 | 21.20 | 89.96 | 25.88 |
| 300 | 1.321 | 21.20 | 89.11 | 24.95 |
| 350 | 1.245 | 21.20 | 87.37 | 23.05 |
| 400 | 1.165 | 21.20 | 85.51 | 21.12 |
| 475 | 1.038 | 21.20 | 82.43 | 18.13 |

from 25.88% at 275 K to 18.13% at 475 K, driven by Voc (1.357 to 1.038 V) and FF while Jsc stays pinned at ∼21.20 mA/cm$^2$, i.e. −0.15%/K relative (−0.039 points/K) over the computed 275–475 K span (−0.16%/K over 300–475 K). The 275 K fill factor (89.96%) sits *below* Green's ideal-diode value at that Voc and temperature (∼91.3%), so no artefact needs invoking; likewise the ceiling's 89.11% is about 1.4 points below ideal (90.5%), leaving genuine FF headroom short of series resistance. The 275 K Voc (1.357 V) sits under the 275 K radiative limit (1.397 V, recomputed at that temperature), so the screen holds across the sweep. The 475 K end of the sweep ignores the temperature dependence of $E_g$ and black-phase stability, so it is a solver stress test, not a prediction.

## 6.1 Parasitic-resistance accounting

Series and shunt resistances are scriptable in SCAPS 3.3.10 (`set external.Rs/Rsh`; `scriptdescription3310.txt`). Computed scripted sweeps at the ceiling base give 24.73% at $R_s = 0.5$ Ωcm$^2$, **24.52%** at $R_s = 1$ Ωcm$^2$, and 24.09% at $R_s = 2$ Ωcm$^2$ (Table 11); a lumped-series analytic derate $\eta = \eta_0/[1 + J_{sc}R_s/(10^3 FF_0 V_{oc})]$—the $10^3$ converting $FF_0V_{oc}$ from V to mV, with $FF_0 = 0.891083$ the 89.11% of Table 6 as a fraction—tracks the computed sweep to within 0.011 points at all three $R_s$ (24.7237/24.5050/24.0790 against 24.7311/24.5160/24.0866). Shunt paths cost 0.14 points at $R_{sh} = 10^4$ Ωcm$^2$ at the prior base; combined $R_s = 1/R_{sh} = 10^4$ at the ceiling gives 24.38%. The ceiling therefore survives parasitics, but only just: at $R_s = 2$ Ωcm$^2$ the margin over the 22.02% certified record is about two points. At routine defects $R_s = 1$ costs 0.40 points (20.82 to 20.42%).

**Table 11:** Best point under computed scripted parasitics (ceiling base; all values computed, none derated).

| Parasitics | PCE (%) | vs 22.02% cert. | Status |
|---|---|---|---|
| $R_s = 0$ (box maximum) | 24.95 | +2.93 | SCAPS box maximum |
| $R_s = 0.5\ \Omega\text{cm}^2$ | 24.73 | +2.71 | optimistic good contact |
| $R_s = 1\ \Omega\text{cm}^2$ | 24.52 | +2.50 | optimistic; ITO sheet resistance is typically higher |
| $R_s = 2\ \Omega\text{cm}^2$ | 24.09 | +2.07 | margin narrowing |
| $R_s = 1/R_{sh} = 10^4$ combined | 24.38 | +2.36 | computed, joint base |

**Table 12:** Pricing the $k_{rad} = 0$ and mobility assumptions (computed scripted runs; $\Delta$ against the $k_{rad} = 0$ anchor of each base). The single-idealisation reference is robust: 20.82% goes to 20.68/19.80% at literature $C_r$; SCAPS's own Eq. 5 value ($3.5 \times 10^{-14}$) is equivalent to omission.

| Run | Voc (V) | Jsc (mA/cm$^2$) | PCE (%) | $\Delta$ (pts) |
|---|---|---|---|---|
| Joint optimum ($k_{rad} = 0$) | 1.321 | 21.20 | 24.95 | – |
| $+C_r = 10^{-10}$ cm$^3$/s | 1.163 | 21.08 | 21.86 | −3.09 |
| $+C_r = 10^{-9}$ cm$^3$/s | 1.113 | 20.43 | 19.83 | −5.12 |
| $\mu_n/\mu_p$ 10/5 | 1.321 | 21.20 | 24.94 | −0.00 |
| $\mu_n/\mu_p$ 5/2 | 1.321 | 21.20 | 24.94 | −0.01 |
| Routine defects ($k_{rad} = 0$) | 1.147 | 20.77 | 20.82 | – |
| $+C_r = 10^{-10}$ cm$^3$/s | 1.141 | 20.73 | 20.68 | −0.14 |
| $+C_r = 10^{-9}$ cm$^3$/s | 1.115 | 20.45 | 19.80 | −1.02 |
| $\mu_n/\mu_p$ 10/5 | 1.148 | 20.60 | 20.50 | −0.32 |

## 6.2 Radiative-recombination and mobility pricing

The survey identifies the zero radiative-recombination default as the primary Shockley–Queisser-violation mechanism: raising $C_r$ from $10^{-10}$ to $10^{-9}$ cm$^3$/s collapses their demo cell from 21.08% to 13.33% [12]. Our campaign sets $k_{rad} = 0$ throughout, so this section prices that choice with scripted `set layer2.radiative` runs at both headline bases (receipts in Table 23; `s23_krad_mu.json`). The script key was validated Appendix-A style: `radiative 0` and `mun/mup 25/25` reproduce the joint and routine receipts digit-identically (24.9464% and 20.8197%), confirming the key paths and the practical units (cm$^3$/s, cm$^2$/Vs); the $10^{-16}$ point reproduces the baseline to 0.0001, bracketing SCAPS's own Eq. 5 value ($3.5 \times 10^{-14}$ cm$^3$/s at our $N_c/N_v$, $E_g$, and thickness), which is therefore equivalent to omitting radiative recombination entirely.

Three findings (Table 12). First, the $k_{rad} = 0$ assumption is worth 3–5 points at the defect floor, almost entirely through Voc (1.321 to 1.163/1.113 V)—the same channel the survey flags, and the reason the ceiling's 0.05 V radiative margin must be read as model-conditional. Second, the cost is regime-dependent like everything else in this study: at routine defects, where SRH recombination dominates, the same $C_r$ values cost only 0.14/1.02 points, so the 20.82% single-idealisation number is robust to the radiative assumption within about a point. Third, the mobility assumption is nearly free at the ceiling (0.00–0.01 points for DFT-grade 10/5 and 5/2 [32] against our 25/25) and small at routine defects (0.32 points, via Jsc and FF): transport is not what holds this cell up, consistent with the interface-led per-decade Morris ranking. One run ($R$_rad1em10) needed a solo retry after a worker crash, noted in the receipts; no result in this table is a crash artefact. A $C_r = 10^{-11}$ bracket point (between omission and literature) costs 1.71 points at the ceiling (23.23%) and 0.01 at routine defects (20.81%): the radiative price is convex in $C_r$ at clean bulk and negligible at routine bulk across the full three-decade bracket. The literature values carry no photon recycling, so this bracket is the pessimistic edge—recycling would return part of the 3–5 points, and the bracket width is the honest statement, not the midpoint. Re-optimising the affinities under radiative loss turns out to be moot: the full 36-point joint map rerun at $C_r = 10^{-11}$ peaks at 23.23% at the same (3.9, 3.7) eV point, and at $C_r = 10^{-10}$ the map compresses to 21.47–21.86% (0.38-point span; the $\chi \geq 3.7$ top cluster sits within

**Table 13:** Stacked all-priced variants (surveyed affinities; $N_t = 10^{15}$, $C_r = 10^{-10}$, $\mu$ 10/5, $E_g = 1.70$, ITO 4.4 eV, $R_s = 1$). An illustrative pessimistic-corner scenario, superseded as headline by the prior-predictive interval of Section 6.5.

| Thickness | Voc (V) | Jsc (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| 0.6 $\mu$m | 1.195 | 18.93 | 85.45 | 19.33 |
| 0.8 $\mu$m | 1.191 | 19.27 | 85.25 | 19.57 |
| 1.0 $\mu$m | 1.188 | 19.38 | 85.09 | 19.59 |
| 1.0 $\mu$m, $R_s = 2$ | 1.188 | 19.38 | 83.60 | 19.25 |

0.03 at 21.83–21.86%), i.e. the radiative channel dominates but residual $\chi$-dependence persists at low $\chi$ (receipts `s30_crchi.jsonl`). The radiative channel dominates the interface activation term rather than shifting it, so the optimum does not move—it dissolves. A routine-$N_t$ subgrid is likewise flat at 20.60%. Pricing $C_r$ at the $k_{rad} = 0$ optimum, as Table 12 does, is therefore the correct accounting, not a shortcut.

The density-of-states choice was challenged as unphysical ($m_h^* \approx 2.2\,m_0$ implied), so it is priced rather than defended: setting $N_v$ to effective-mass physical values ($4.1 \times 10^{18}$/$8.9 \times 10^{18}$ cm$^{-3}$ for $m^* = 0.3/0.5\,m_0$; script path verified digit-identical first) *raises* the ceiling to 26.50/26.11% (Voc 1.390/1.373 V) and the routine case to 22.25/21.91%. Physical DOS inflates, capped DOS suppresses—so the capped values run conservative, and every headline in this paper is the lower choice. Crucially, the radiative bound does *not* move with DOS: it depends on absorptance and $E_g$ only, so the 1.390 V physical-DOS ceiling run *fails* the 1.371 V screen and is excluded by it—reported here, not headlined. (The $\sim$0.08 V figure sometimes quoted is $kT\ln 20$ from the 20-fold fall in $n_i^2$, i.e. $n_i$ falling $\sim$4.5-fold; that shift applies to a diode with a radiative channel, which this model omits.) The cap therefore stands as a conservative stand-in for the missing radiative channel, kept for continuity across the campaign, with physical DOS retained as a sensitivity; adding a radiative channel ($C_r = 10^{-10}$) to the physical-DOS ceiling case brings Voc back under the bound (1.241/1.221 V at 23.46/23.04%; receipts `s31_small.json`), making the screen self-consistent exactly as predicted. Reproducing the source paper's printed DOS instead ($N_c = 1.1 \times 10^{20}$/$N_v = 8.2 \times 10^{20}$) costs 2.23 points at the baseline (18.99 to 16.76%, Voc 1.122 to 1.005 V) and 2.36 at routine defects (20.82 to 18.46%): the baseline replication depends on the DOS cap by an amount comparable to the interface effects under study, and is reported with it, not without it.

## 6.3 Stacked realistic variants

Single-assumption prices interact, so the honest headline is a stacked case, not a sum of discounts. Four variants combine the routine-defect base with literature assumptions simultaneously—$N_t = 10^{15}$ cm$^{-3}$, $C_r = 10^{-10}$ cm$^3$/s, $\mu_n/\mu_p = 10/5$, $E_g = 1.70$ eV with linked $\chi = 3.953$ eV, ITO 4.4 eV, $R_s = 1\ \Omega$cm$^2$ (receipts: `s24_stack.json`)—at absorber thicknesses spanning the experimental film range (Table 13).

The stacked range is 19.25–19.59%: below the 22.02% certified record, below the 20.82% single-idealisation case by about a point, and within reach of the best measured CsPbI3 devices rather than above them. Thickness structure is flat across the experimental window (0.6–1.0 $\mu$m within 0.26 points), and the $R_s = 1 \to 2$ step costs 0.34 points on top. Two readings follow. First, the single-assumption prices of Tables 12 and 20 do not add linearly, but they also do not cancel: stacking six literature-grade corrections moves the headline about 1.2 points, dominated by the contact and radiative terms. Second, with literature $C_r$ alone the ceiling falls to 21.86% (Table 12)—below the certified record without any other correction. The 24.95% ceiling exceeds the record only because it requires *every* idealisation simultaneously; remove any load-bearing one and it leaves the record behind.

**Table 14:** Calibration ladder at measured-grade inputs ($E_g = 1.70$, ITO 4.2 eV, $R_s = 1$, $C_r = 10^{-10}$, $\mu$ 10/5, $N_{if} = 3 \times 10^{10}$ unless noted). One $3 \times 10^{14}$ retry after a worker crash; receipts in Table 23.

| Case | Voc (V) | Jsc (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| 0.6 $\mu$m, $N_t = 10^{14}$ | 1.203 | 18.91 | 87.00 | 19.79 |
| 0.8 $\mu$m, $N_t = 10^{15}$ | 1.185 | 19.14 | 85.18 | 19.33 |
| 0.8 $\mu$m, $N_t = 3 \times 10^{14}$ | 1.198 | 19.26 | 86.47 | 19.94 |
| 0.8 $\mu$m, $N_t = 10^{14}$ | 1.203 | 19.29 | 86.90 | 20.16 |
| 0.8 $\mu$m, $N_t = 10^{13}$ | 1.205 | 19.31 | 87.13 | 20.27 |
| 1.0 $\mu$m, $N_t = 10^{14}$ | 1.202 | 19.47 | 86.83 | 20.32 |
| 0.8 $\mu$m, $N_t = 10^{14}$, $N_{if} = 10^{10}$ | 1.216 | 19.43 | 87.42 | 20.66 |

## 6.4 Calibration against measured devices

A realistic-projection claim should be tested against a real device, so we drove the model toward measured-grade inputs: $N_t = 10^{14}$ cm$^{-3}$, $N_{if} = 3 \times 10^{10}$ cm$^{-2}$, $E_g = 1.70$ eV with linked affinity, $R_s = 1$ $\Omega$cm$^2$, ITO 4.2 eV, DFT mobilities, $C_r = 10^{-10}$ cm$^3$/s (receipts: `s28_calib.json`; ITO 4.2 eV via a new archived definition variant). The ladder climbs monotonically and saturates (Table 14): 19.33% at $N_t = 10^{15}$, 19.94% at $3 \times 10^{14}$, 20.16% at $10^{14}$, 20.27% at $10^{13}$ (0.8 $\mu$m throughout), reaching 20.32% at 1.0 $\mu$m and 20.66% with interfaces at the $10^{10}$ floor (Voc 1.216 V).

This is a partial calibration, reported as such: the model with literature-grade inputs climbs to 20.3–20.7% at Voc 1.20–1.22 V, approaching but not reproducing the 22.02% certified record at 1.27+ V. The remaining $\sim$1.5 points and $\sim$60 mV mark the distance between routine films and record films, and closing them requires cleaner-than-routine bulk ($N_t < 10^{13}$ cm$^{-3}$) or better-than-floor interfaces—exactly the experimental frontier this framework cannot cross by itself. The calibration bounds the model in both directions: it is not pessimistic noise (it climbs monotonically toward the records as inputs clean), and it is not a record predictor (it saturates well below them at defensible inputs).

## 6.5 Prior-predictive interval

The stacked range of Section 6.3 is a pessimistic corner, not an uncertainty range: it varies thickness plus one $R_s$ step at a single stacked assumption set. The replacement samples the prior volume directly. With the frozen priors of `priors.md` (log $N_t$ 13–16, log $N_{if}$ 10–13, log $C_r$ $-11$–$-9$, mobility scale 0.2–1, $R_s$ 0.5–3, categorical ITO/Ni work functions, thickness 0.4–1.5 $\mu$m; surveyed affinities held fixed to avoid the bimodal contact cliffs), a 96-point Latin hypercube (seed 11) plus an 8-run $C_r = 0$ sensitivity subset gives, at 103 converged runs with one genuine divergence (Q002: $N_t = 10^{15.57}$, ITO 4.4 eV with Ni 5.0 eV—both contacts at their edges—hangs the solver repeatedly and is recorded as non-convergent, not excluded), a prior-predictive median of **18.0%** with a 5–95% interval of **15.6–20.0%** (20k-bootstrap CIs: median [17.7, 18.1], p5 [15.3, 16.1], p95 [19.7, 20.2]; receipts `s29_lhs.jsonl`, `s29_analysis.json`). Zero runs trip the per-run Voc (1.418 V at 1.70 eV) or Green-FF screens. Rank importances (Spearman): $N_{if}$ $-0.78$, $N_t$ $-0.37$, thickness $+0.26$, mobility $+0.21$, $C_r$ $-0.19$, $R_s$/ITO/Ni near zero. The $C_r = 0$ sensitivity median is 17.9%, so the radiative default barely moves the prior-volume median—its bite is concentrated at the clean corner. Propagating the $\pm$1.2 mA optical scale post-hoc widens absolute readings by $\sim$1 point; the interval as quoted is electrical-only. Adding effective-mass density of states as a ninth prior dimension (log $N_v$ 18.6–19.9, seeds 13/14; amendment in `priors.md`) moves the interval almost not at all: median 18.2% [17.8, 18.5], 5–95% 15.8–20.0%, at 104 converged runs with zero screen flags (receipts `s32_lhs_dos.jsonl`, `s32_analysis.json`). The DOS dimension ranks near last in-volume (Spearman $-0.04$) even though physical DOS swings the clean corner by $+1.5$ points—corner sensitivity without volume sensitivity, which is why the interval is reported both ways and both agree.

Two readings are stated plainly. First, this is a prior-predictive interval under stated priors, not a

confidence interval on real devices: change the priors and the interval moves, and Table 22 carries them so it can. Second, every measured record in Table 1 sits above the 95th percentile—the envelope contains routine films and the stacked range, not record films, and the gap quantifies what record quality demands beyond these priors rather than validating against them.

## 7.0 Interface dose–response and the limits of auditing others

Deleting the interface blocks (audit pairs in Table 17, dose–response plotted in Figure 10) inflates efficiency by 1.26–1.66 points absolute at the surveyed base (23.40 to 24.87%, 23.47 to 24.73%, 23.50 to 25.16%), almost entirely through a 0.13 V Voc rise as interface recombination vanishes. But the binary ON/OFF comparison understates the audit: the full dose–response at the prior adopted base (Table 15) runs 25.11% at $10^8$ cm$^{-2}$, 24.77% at $10^9$ cm$^{-2}$, 23.84% at the $10^{10}$ cm$^{-2}$ guardrail, 22.62% at $10^{11}$ cm$^{-2}$, and 21.20% at $10^{12}$ cm$^{-2}$. Because interface density and cross-section enter only as the product $N\sigma$, Table 16 restates the guardrailed regime in the portable metrics other studies can compare against: bulk lifetime $\tau = 1/(\sigma v_{th} N_t)$ and interface velocity $S = \sigma v_{th} N_{if}$ (with $v_{th} = 10^5$ m/s). Our guardrail is $\tau = 100$ $\mu$s with $S = 100$ cm/s. The decisive screen, however, is energetic, not parametric: the detailed-balance Voc computed from the model's own Eg-sqrt absorption is 1.371 V at 1.65 eV (1.418 V at 1.70 eV), and the no-interface cells we computed (1.363–1.382 V) sit within 0.011 V of that radiative limit—the top one above it, the rest with no margin either—because with $k_{rad} = 0$ in the model there is nothing to hold the quasi-Fermi splitting below the radiative bound. A band that hugs the limit is not evidence of a working cell. The published 24.17% [8] and 24.24% [9] values are therefore not placed on our curve at all: their stacks, gaps, thicknesses and defect parameters all differ from ours, neither paper prints a complete parameter table from which $\tau$ or $S$ could be reconstructed, and we state this as a limitation rather than a finding.

### 7.1 Where the missing points go

The inflation is almost entirely a Voc effect (+0.13 V at the ceiling cell, +0.148 V at 1.5 $\mu$m) with Jsc unchanged to within 0.02 mA/cm$^2$: removing the $10^{10}$ cm$^{-2}$ interface states eliminates the dominant SRH channel at the heterojunctions, unpinning the quasi-Fermi splitting. The no-interface Voc values (1.363–1.382 V) all sit within 0.011 V of the 1.371 V detailed-balance limit computed from the model's own absorption, the top one above it: with $k_{rad} = 0$ there is no radiative channel to pin the splitting, so none of these cells carries margin and none is trusted, regardless of rank order. Fill factor moves non-monotonically (89.86% down to 85.25–87.05% across the three audited cells) as the recombination profile reshapes. The no-interface 87.05% at Voc 1.363 V sits 3.7 points below Green's ideal (90.8%): the archived JV curve stays flat to 1.16 V with no kink or S-shape, so there is no visible extraction barrier, but the knee remains SRH-rounded by the surviving bulk channel—removing interfaces unpins Voc without idealising the diode. This bounds the "no extraction barriers" claim to current collection, not to fill factor. Asymmetric tests show both heterojunctions matter: $10^{12}$ cm$^{-2}$ on the ETL side alone gives 21.35%, on the HTL side alone 21.20%. Raising the capture cross-section 10$\times$ at fixed $N_{if} = 10^{10}$ cm$^{-2}$ gives 22.2108%, digit-identical to the $N \times 10$ run at the same base (both 22.2108; receipts in Table 23): for the symmetric single level, rate scales exactly with the $N\sigma$ product. The same identity holds in the bulk: $\sigma \times 10$ at the $N_t$ floor gives 23.2572% (digit-identical to $N_t \times 10$ at the same base, = 23.2572; receipts in Table 23), a 0.25-point cost against the surveyed 23.5041 anchor. (A first attempt via the script `set` key for $\sigma$ neutralised the interfaces instead of scaling them; only the verified definition-file variant is reported—see Appendix A.) Bulk $N_t \times \sigma$ costs 0.25 points at the $10^{12}$ cm$^{-3}$ floor but 3.12 points at routine $10^{15}$ cm$^{-3}$ (20.56% at matched $\sigma \times 1$ geometry down to 17.44%), while $\sigma \times 0.1$ at the floor gains only 0.03 points (23.53%)—the floor is saturated.) The computed parasitic sweep at the adopted base is given in Table 11.

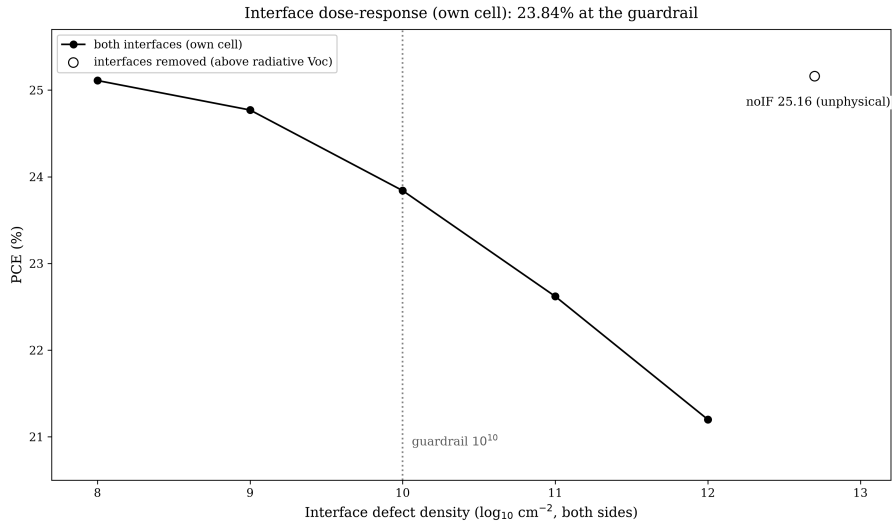



**Figure 10:** Same cell at graded interface quality: efficiency vs $N_{if}$ (both sides). The no-interface point is placed off-scale right (12.7) by construction. No literature scale bar is shown: cross-stack mapping is not claimed (Section 7.0).

**Table 15:** Interface dose–response (prior adopted base unless noted; both sides unless noted).

| $N_{if}$ (cm$^{-2}$) | PCE (%) | Note |
|---|---|---|
| $10^8$ | 25.11 | – |
| $10^9$ | 24.77 | – |
| $10^{10}$ (guardrail) | 23.84 | prior adopted base |
| $10^{11}$ | 22.62 | – |
| $10^{12}$ | 21.20 | – |
| $10^{12}$ ETL side only | 21.35 | surveyed base |
| $10^{12}$ HTL side only | 21.20 | surveyed base |
| $10^{10}$, $\sigma \times 10$ (verified) | 22.21 | matches $N \times 10$ (rate $\propto N\sigma$) |
| removed (noIF) | 25.16 | above detailed-balance Voc: unphysical |

**Table 16:** Guardrailed regime in portable metrics ($\sigma = 10^{-19}$ m$^2$, $v_{th} = 10^5$ m/s): bulk lifetime $\tau$ and interface velocity $S$.

| Density | $\tau$ / $S$ | Note |
|---|---|---|
| $N_t = 10^{12}$ cm$^{-3}$ (floor) | $\tau = 100$ $\mu$s | 1–2 orders beyond best films |
| $N_t = 10^{13}$ cm$^{-3}$ | $\tau = 10$ $\mu$s | – |
| $N_t = 10^{14}$ cm$^{-3}$ | $\tau = 1$ $\mu$s | – |
| $N_t = 10^{15}$ cm$^{-3}$ (routine) | $\tau = 0.1$ $\mu$s | routine case |
| $N_{if} = 10^{10}$ cm$^{-2}$ (guardrail) | $S = 100$ cm/s | floor of stated box |
| $N_{if} = 10^{11}$ cm$^{-2}$ | $S = 1000$ cm/s | – |

**Table 17:** Guardrail audit: identical cells with interfaces ON versus OFF.

| Cell | IF ON (%) | IF OFF (%) | Inflation (pts) |
|---|---|---|---|
| 1.5 $\mu$m cell | 23.40 | 24.87 | +1.47 |
| 2.0 $\mu$m cell | 23.47 | 24.73 | +1.26 |
| Surveyed-base best | 23.50 | 25.16 | +1.66 |

## 8.0 Morris sensitivity screening

Grid methods with uneven ranges cannot rank physical importance, so we ran a Morris sensitivity screen instead: $r = 8$ trajectories, $p = 4$ levels, $k = 6$ factors (absorber thickness, $\log N_A$, $\log N_t$, $\log N_{D,ETL}$, $E_g$ with the $\chi$ link applied deterministically, and $\log N_{if}$ tied on both interfaces), 56 fresh simulations, all converged, spanning 15.34–22.96% (Table 18, Figure 11). The interface range is $10^{10}$–$10^{12}$ cm$^{-2}$ throughout—inside the stated box, unlike a precursor screen (seed 7, $10^{8}$–$10^{12}$) whose stronger interface lead was computed partly in banned territory and is superseded. Per decade, interface quality leads at 0.85 points (bootstrap 95% CI [0.42, 1.35], 20000 resamples) ahead of bulk defect density at 0.37 (CI [0.15, 0.60]). Because marginal intervals overlap, we ran the sharper paired test the design allows: the per-trajectory difference $|EE_{Nif}| - |EE_{Nt}|$ averages +0.58 points with a bootstrap 95% CI of [-0.59, 1.97] (5 of 8 trajectories positive)—inside the guardrailed box, the interface lead is *not* significant by Morris alone, and the regime claim rests on the dose–response and the $S$-vs-$d/\tau$ ratio instead. Per eV, bandgap moves 22.6 points; per $\mu$m, thickness 0.83. These per-unit numbers are not commensurable across units, the interface figure bundles both heterojunctions over a 2-decade range against 3 for bulk, and—crucially—the $E_g$ factor confounds gap with band offset, since the $\chi$ link assigns half of every gap step to the conduction band while Section 5.0 shows alignment alone moves efficiency by several points. The gap-step pattern betrays the confounding (all on the prior adopted base, 23.8365 anchor): 1.65 to 1.68 eV costs 1.13 points but 1.68 to 1.70 eV only 0.19 (and 1.70 to 1.72 eV costs 0.93), tracking the $E_a$ landscape rather than the absorption edge. On the ceiling base the same steps even change sign (24.44 to 24.73% from 1.68 to 1.70 eV), confirming the steps map the activation landscape, not the gap. The decisive test remains the dose–response repeated at routine defects: at $N_t = 10^{15}$ cm$^{-3}$ and 1.0 $\mu$m, $N_{if}$ $10^{9}$–$10^{11}$ cm$^{-2}$ moves efficiency only from 20.84% to 20.67% (0.16 points over two decades), while the $N_t$ $10^{14}$–$10^{15}$ step costs far more. At the defect floor interfaces dominate; at routine defects bulk defects do. The regime dependence has a one-ratio explanation, qualified carefully: compare the *nominal* interface velocity $S = \sigma v_{th} N_{if}$ with the effective bulk velocity $d/\tau$. At the ceiling, $d/\tau \approx 2.2$ cm/s (2.2 $\mu$m over 100 $\mu$s) sits far below $S = 100$ cm/s, so heterojunction recombination gates the voltage; at routine defects, $d/\tau \approx 1000$ cm/s (1.0 $\mu$m over 0.1 $\mu$s) dwarfs $S$, so bulk SRH does. This is the textbook effective-lifetime form $1/\tau_{\text{eff}} = 1/\tau_b + 2S/d$ for two interfaces [33], not a new criterion—and nominal $S$ is a transferable *metric*, not a transferable *prediction*: at fixed $N_{if}$ and $\sigma$, efficiency moves from 22.1% to 24.95% across the $\chi_{\text{CBTS}}$ sweep because band alignment sets the interface carrier densities, so the same nominal $S$ recombines differently. The Morris ranking, the dose–response flattening, and the $C_r$ cost collapse (Section 6.2) are three views of this single ratio crossing unity, each conditional on the alignment it was computed in. Binning the 95 converged main-batch runs by $2S\tau/d$ finds no clean crossing at this sample size (both factors carry $|\rho| \sim 0.7$–$0.9$ in every bin; receipt `s29_analysis.json`): the ratio organises the two anchor regimes, it does not resolve the ranking bin-by-bin inside the prior volume. On the screen's own history: a precursor design (seed 7) sampled $N_{if}$ down to $10^{8}$ cm$^{-2}$ and found a stronger interface lead; because that range breaches the stated floor, the precursor is superseded by the guardrail-clean rerun reported here, and only the rerun enters the headline ranking. The $n = 8$ bootstrap intervals should be read as lower bounds on uncertainty. Trap depth is a non-factor at this defect level: moving $E_t$ to 0.3 or 0.9 eV changes efficiency by at most 0.012 points (23.8259% and 23.8241% against 23.8365%).

The regime-dependent interface role is consistent with passivation studies in which interface-adjacent defects gate the voltage [34], and with Bayesian optimisation on CsSnI3 where the optimal thickness likewise shifts with defect density [10].

## 9.0 Discussion

Fabrication plausibility bounds the result, and the computed sweeps price each term. $N_A = 3 \times 10^{16}$ cm$^{-3}$ is a modeling choice spanning reported simulation ranges; measured doping densities in

**Table 18:** Morris screen design and results ($r = 8$ trajectories, $p = 4$, $\Delta = 2/3$, seed 21; 56/56 converged; guardrail-clean $N_{if}$ range). $\mu^*$ and $\sigma$ are per normalised unit input; the per-decade column (sixth) and the CI column (seventh) are both per physical decade (per eV or per $\mu$m where noted); $n = 8$ unique directed effects per factor; CI = bootstrap 95% interval (20000 resamples), also per physical decade.

| Factor | Range | $\mu^*$ | $\sigma$ | $n$ | per decade | 95% CI |
|---|---|---|---|---|---|---|
| $\log N_{if}$ (both IF) | $10^{10}$–$10^{12}$ cm$^{-2}$ | 1.70 | 1.35 | 8 | 0.85 | [0.42, 1.35] |
| $\log N_t$ (absorber) | $10^{12}$–$10^{15}$ cm$^{-3}$ | 1.12 | 0.97 | 8 | 0.37 | [0.15, 0.60] |
| $E_g$ (+$\chi$ link) | 1.65–1.75 eV | 2.26 | 1.27 | 8 | 22.6/eV | [14.1, 31.7] |
| Thickness | 0.4–2.2 $\mu$m | 1.50 | 1.27 | 8 | 0.83/$\mu$m | [0.36, 1.33] |
| $\log N_{D,ETL}$ | $10^{16}$–$10^{19}$ cm$^{-3}$ | 0.24 | 0.19 | 8 | 0.08 | [0.04, 0.12] |
| $\log N_A$ (absorber) | $10^{15}$–$3\times10^{16}$ cm$^{-3}$ | 1.11 | 1.24 | 8 | 0.74 | [0.39, 1.15] |

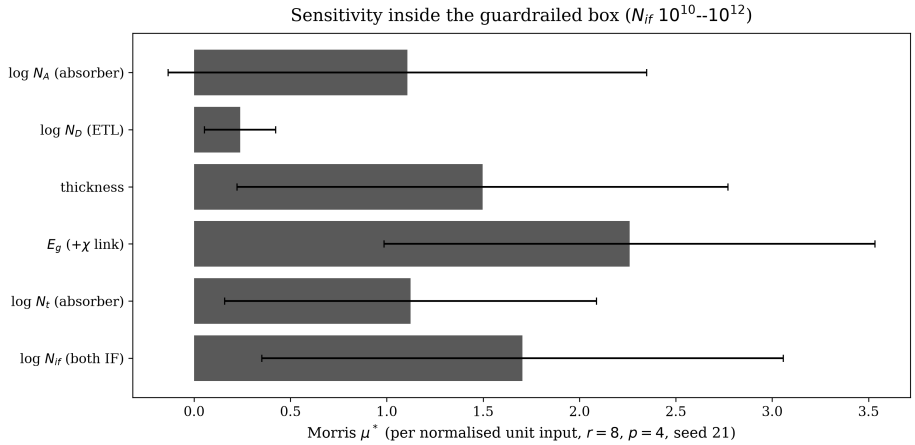



**Figure 11:** Morris $\mu^*$ (bars) with interaction spread $\sigma$ (whiskers). Per-decade, per-eV, and per-$\mu$m figures are not commensurable across units; ranking is regime-conditional: interfaces lead at the defect floor, bulk defects at routine levels (Table 20); large $\sigma$ signals interactions.

**Table 19:** The reference routine-defect case, defined once: 1.0 $\mu$m absorber, $N_A = 3\times10^{16}$ cm$^{-3}$, $N_t = 10^{15}$ cm$^{-3}$, $N_{D,\mathrm{ETL}} = 9\times10^{17}$ cm$^{-3}$, $E_g = 1.65$ eV/$\chi = 3.928$ eV, surveyed affinities ($\chi_{\mathrm{CBTS}} = 3.6$/$\chi_{\mathrm{TiO2}} = 4.0$ eV), interfaces at $10^{10}$ cm$^{-2}$, Ni 5.5 eV / ITO 4.0 eV, $R_s = 0$: 20.82%. The 20.86% figure elsewhere is the joint-affinity-base thickness-factorial peak at the same $N_t$, not this case.

| Parameter | Value |
|---|---|
| Absorber thickness / $N_A$ / $N_t$ | 1.0 $\mu$m / $3\times10^{16}$ / $10^{15}$ cm$^{-3}$ |
| ETL $N_D$ / gap / affinity | $9\times10^{17}$ cm$^{-3}$ / 1.65 eV / 3.928 eV |
| Affinities / contacts | surveyed $\chi$; Ni 5.5 / ITO 4.0 eV |
| Interfaces / parasitics | $10^{10}$ cm$^{-2}$ both sides; $R_s = 0$, $k_{rad} = 0$ |
| Result | 20.82% (20.42% at scripted $R_s = 1$) |

**Table 20:** Routine-defect case (Table 19): interface dose–response is nearly flat, so bulk defects dominate here. Dose rows are 1.0 $\mu$m and $\Delta$ is against the 20.82% guardrail row of that base; the two ITO rows are 0.8 $\mu$m runs (20.72% at surveyed $\chi$) and their $\Delta$ is against that 0.8 $\mu$m row, not the 1.0 $\mu$m one. Values computed; $R_s = 0$ unless stated.

| $N_{if}$ (cm$^{-2}$) | PCE (%) | $\Delta$ vs guardrail | Base |
|---|---|---|---|
| $10^9$ | 20.84 | +0.017 | surveyed $\chi$ |
| $10^{10}$ (guardrail) | 20.82 | – | surveyed $\chi$ |
| $10^{11}$ | 20.67 | −0.147 | surveyed $\chi$ |
| $10^{10}$, ITO 4.4 eV (0.8 $\mu$m) | 20.68 | −0.04 | surveyed $\chi$ |
| $10^{10}$, ITO 4.4 eV (0.8 $\mu$m) | 18.11 | −2.61 | adopted $\chi$ |
| $10^{10}$, $R_s = 1$ $\Omega$cm$^2$ | 20.42 | −0.40 | surveyed $\chi$; computed (s14) |

lead-halide perovskites sit below $10^{12}$ cm$^{-3}$ [35], so this ceiling is optimistic and its price is carried by the Morris $\log N_A$ factor; $E_g = 1.65$ eV sits below the $\sim$1.7 eV of the experimental records; its contribution is base-dependent—roughly one point on the surveyed base (best 1.694 eV grid point: 22.506%), but only 0.22 points on the ceiling base (24.73% at 1.70 eV), where the gap steps map the $E_a$ landscape rather than absorption (Section 8.0). At 1.70 eV the model still exceeds the certified record by about 2.7 points, so the excess cannot be laid on the gap: it comes mostly from fill factor ($\sim$89% simulated against $\sim$83% measured) plus the permissive contacts and affinities priced above; $N_t = 10^{12}$ cm$^{-3}$ demands beyond-best-film quality ($\tau_{\mathrm{SRH}} \approx 100$ $\mu$s, Table 16). The contacts are load-bearing. At the surveyed anchor Ni 5.5 to 5.2 eV costs nothing (23.5041 to 23.5039%, 0.0002 apart and inside the resolution floor) while Ni 5.0 eV costs 0.25 points (23.2571%); on the ceiling affinities the same Ni 5.0 eV choice instead collapses the cell to 15.63% (Voc 1.01 V) against 18.83% at Ni 5.2 eV—a back-contact barrier cliff, not a slope, so the back-contact tolerance window collapses as the bands shift with affinity. ITO 4.0 to 4.7 eV collapses the cell to 18.60% (Voc 1.08 V)—SCAPS's thermionic-only contact boundary, so 4.0 eV is the edge of what this model tolerates rather than a typical ITO value; the routine-defect penalty is larger still, with ITO 4.4 eV costing 0.04 points at surveyed $\chi$ (20.72 to 20.68% on the 0.8 $\mu$m base) but 2.61 points at adopted $\chi$ (20.72 to 18.11% on the same base): the front-contact tolerance window itself shifts with ETL affinity (Table 20). Both optimum affinities are aggressive ($\chi_{\mathrm{TiO2}} = 3.7$ below the typical 4.0–4.2 eV window; $\chi_{\mathrm{CBTS}} = 3.9$ above the surveyed 3.6 eV) and move efficiency through the interface activation energy rather than transport (Table 7). At routine settings the model gives 20.82% (1.0 $\mu$m, $N_t = 10^{15}$ cm$^{-3}$, surveyed $\chi$), 20.68% with ITO at 4.4 eV on the 0.8 $\mu$m base, 18.11% when that ITO is combined with the adopted $\chi_{\mathrm{TiO2}}$, and 20.42% with $R_s = 1$ $\Omega$cm$^2$. Against experiment (Table 21): 24.95% exceeds the 22.02% certified record by 2.93 points; the excess over the best measured devices comes mostly from fill factor ($\sim$89% simulated against $\sim$83% measured) plus permissive contacts and affinities, with voltage contributing its $\times$1.02 share and the gap contributing 0.22 points on the joint base—and we do not attribute that FF excess to any single cause: the model carries no transport-layer bulk recombination and no back reflector, two systematic choices that both bias FF upward by amounts this study does not isolate, while the 0.017-point IV-step uncertainty only sets the significance threshold for any of these comparisons—our Voc deficit (0.33 V against $E_g/q$ at 1.65 eV) is narrower than the records' $\sim$0.41 V against $\sim$1.70 eV: against the 22.16% device the excess splits as FF$\times$1.07, Voc$\times$1.02, Jsc$\times$1.02, so voltage contributes its share and the balance sits in fill factor plus contacts. With computed $R_s = 1$ $\Omega$cm$^2$ (24.52%) the margin is 2.50 points, and at attainable gaps on the ceiling base the model gives 24.44% ($E_g = 1.68$ eV), 24.73% ($E_g = 1.70$ eV) and 24.28% ($E_g = 1.72$ eV), with 24.34% at $R_s = 1$ $\Omega$cm$^2$ ($E_g = 1.70$ eV; $\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$, ITO 4.0 eV throughout). The non-monotonic 1.68-to-1.70 step (+0.29 points) is the $E_a$ confounding of Section 8.0, not an absorption effect. The result sits at about four-fifths of the $\sim$30–31% Shockley–Queisser limit at 1.65 eV. A final calibration note: integrating the computed QE against the AM1.5G spectrum gives 22.39 mA/cm$^2$ against the 21.20 mA/cm$^2$ IV value—a 1.19 mA/cm$^2$ gap we carry as a two-sided model-form term (headline Jsc uses the converged IV value) alongside the $\sim$1 mA/cm$^2$ UV enve-

**Table 21:** Loss budget: permissive-box ceiling against measured record devices and model limits. FF is where the simulation exceeds experiment, not Voc. PCE values are each paper's headline; the Lei row is a mixed-source triple (its printed FF gives 22.24%, its reported headline 22.16%).

| Device | Voc (V) | Jsc (mA/cm$^2$) | FF (%) | PCE (%) |
|---|---|---|---|---|
| This work (box ceiling) | 1.321 | 21.20 | 89.11 | 24.95 |
| This work + $C_r = 10^{-10}$ (scripted) | 1.163 | 21.08 | 89.13 | 21.86 |
| This work, stacked realistic range | 1.19 | 18.9–19.4 | 83.6–85.5 | 19.25–19.59 |
| AP-modified CsPbI3 [37] | 1.29 | 20.77 | 83.01 | 22.16 |
| QD-interface CsPbI3 [38] | 1.28 | – | – | 22.15 |
| Dual-interface CsPbI3 [4] | 1.27 | – | – | 21.71 |
| Carbon HTL-free [39] (cert. 20.35) | 1.202 | 20.72 | 83.2 | 20.72 |
| Radiative limit (ideal, step absorption) | 1.372 | 24.4 (SQ) | 90.5 | – |

lope.

The framework bites on live claims. A 2026 machine-learning-coupled screen reports 30.78% for a $Sr_3PBr_3$ absorber with a CBTS hole transport layer—our HTL family—at Voc 1.32 V, FF 87.05%, with SHAP importances led by acceptor and defect densities [36]. Three checklist rows fail on the published record: no interface status or dose is stated, the radiative coefficient is not reported, and the 30.78% headline sits above the ∼26–27% single-junction ceiling for its gap class (SQ Jsc 19.8 at 1.8 eV with Voc ∼1.45–1.5 V and FF 0.9). Our audit suggests the direction of each correction (interfaces on, literature $C_r$, scripted parasitics), all pointing down—but those magnitudes are measured on CsPbI3 and are not transferable to $Sr_3PBr_3$, so no corrected number is quoted. A model-independent check sharpens the audit without touching their stack: their quoted Voc (1.32 V), FF (87.05%), and PCE (30.78%) imply Jsc $= 30.78/(1.32 \times 0.8705) = 26.8$ mA/cm$^2$ (their quoted 26.82 confirms the arithmetic), against an ideal step-absorption SQ Jsc of 19.8 mA/cm$^2$ at 1.8 eV (26.3 even at 1.6 eV)—a breach for any gap at or above ∼1.6 eV. Their absorber gap is not stated in the records accessible to us, so the conditional stands: either the current breaches the radiative ceiling, or the gap sits below the wide-gap rationale of a lead-free top cell. We do not re-simulate their stack—different absorber, different gaps—so this is a reporting audit, not a recomputation; the point is that the checklist adjudicates a published 30%+ claim in one paragraph, which is what a standard is for.

## 10.0 Limitations and outlook

A laboratory translation would need, simultaneously: ∼0.4–1.5 $\mu$m CsPbI3 films spanning the 15.6–20.0% prior-predictive interval (Section 6.5; time-resolved photoluminescence verified), interface passivation toward $10^{10}$ cm$^{-2}$ following the fluorinated-dipole [2] or dual-interface [4] chemistries, a front contact near 4.0 eV, and $R_s \leq 1$ Ωcm$^2$ (Table 11). Not demonstrated here: 2.2 $\mu$m device-quality CsPbI3 at $N_t \sim 10^{12}$ cm$^{-3}$, and sputtered CBTS directly on CsPbI3 (CBTS typically requires high-temperature sulfurisation), for which we found no precedent—so this section is a requirements list, not a recipe. In-line controls, where applicable: QE edge at 750 nm (Figure 5) and the −0.15%/K temperature coefficient (Table 10).

Five boundaries delimit this work. First, $N_t = 10^{12}$ cm$^{-3}$ at 2.2 $\mu$m demands films cleaner than the best reported, and with the modelled $\sigma = 10^{-19}$ m$^2$ implies $\tau_{\mathrm{SRH}} \approx 100$ $\mu$s; a reader preferring $\sigma = 10^{-17}$ m$^2$ would get 1 $\mu$s, so the lifetime claim scales as $1/\sigma$ and the defect density, not the lifetime, is what we hold fixed; at routine $10^{15}$ cm$^{-3}$ a dedicated sweep peaks at 20.82% (1.0 $\mu$m), 20.68% with ITO at 4.4 eV on the 0.8 $\mu$m base, 18.11% when that ITO meets the adopted $\chi_{\mathrm{TiO2}}$, and 20.42% with $R_s = 1$ Ωcm$^2$. Second, the contacts (Ni 5.5 eV, ITO 4.0 eV), the joint-optimum affinities ($\chi_{\mathrm{CBTS}} = 3.9/\chi_{\mathrm{TiO2}} = 3.7$ eV, both outside surveyed/typical ranges), and the barrier-side hole offset (+0.22 eV) are permissive-edge stated-box choices, each priced in Section 9.0. Third, phase

**Table 22:** Guardrailed-reporting checklist: the minimum disclosure for a comparable SCAPS claim. The Protocol column maps each row to the standard simulation protocol of [12] (realistic inputs; full reporting; effective spectra and absorption; SQ check; explained deviations), whose optical half is implemented in [13].

| Item | Disclose | This work | Protocol |
|---|---|---|---|
| Interfaces | status + $N_{if}$ + $\sigma$ (or $S$) | $10^{10}$ cm$^{-2}$, $10^{-19}$ m$^2$ ($S$ = 100 cm/s) | reporting |
| Bulk defects | $N_t$ + $\sigma$ (or $\tau$) | $10^{12}$ cm$^{-3}$ (100 $\mu$s); routine $10^{15}$ | reporting |
| Radiative coeff. | $C_r$ value used | 0, priced at $10^{-10}$/$10^{-9}$ (Sec. 6.2) | inputs |
| Mobilities | $\mu_n/\mu_p$ + anchor | 25/25, priced at 10/5, 5/2 (Sec. 6.2) | inputs |
| Units | script vs file systems | Appendix A (two traps) | reporting |
| Parasitics | $R_s/R_{sh}$ basis | scripted; routine case incl. | inputs |
| Contacts | both work functions | Ni 5.5 / ITO 4.0 eV, priced | inputs |
| Affinities | surveyed vs adopted $\chi$ | joint optimum, priced ($\chi$-link) | no isolated opt. |
| Optics | $R$, absorption, spectrum | $R = 10\%$ flat; edge-moving Eg-sqrt; Jsc uncert. carried; TMM spectra *not implemented* | spectra |
| SQ screen | Voc limit + deviations | 1.371 V at 1.65 eV; no-IF disqualified | SQ check |
| Receipts | defs + raw outputs | public repo (no request step) | reporting |

stability of black CsPbI3 against the yellow $\delta$-phase, ion migration, $E_g(T)$, and damp-heat degradation are outside SCAPS-1D steady-state scope and must be validated experimentally, following the passivation routes of [1, 2, 4]. Fourth, the grid is a screen plus a 104-configuration optimiser, not a proven global optimum: the ceiling sits at the corner of the allowed box, the exploratory search peaks 1.87 points below it among guardrail-satisfying runs, and the leading cluster is numerically inseparable. Fifth, the single-junction ceiling ($\sim$30% at 1.65 eV [27]) caps how far any interface-inclusive screening can go.

## 11.0 Conclusion

Grid screening under interface-inclusive guardrails finds a 24.95%-efficient permissive-box ceiling in the stated CsPbI3/CBTS box and a prior-predictive median of 18.0% at routine defect levels, with every load-bearing assumption priced by computed sweep: beyond-best-film defect levels, a low gap, a work-function-tuned front contact, joint affinity optimisation at aggressive values, and—new against the literature—the radiative-recombination default and the mobility values. Designed as the electrical-loss companion to the optical-loss framework of [13] and a partial implementation of the standard simulation protocol of [12] (four of five items; the optical half is cited), this study maps each known overestimation cause to one priced row. An exploratory search over a wider box found nothing higher, though by construction it does not bracket the ceiling. The same campaign prices the routine-defect thickness optimum, shows the interface dose–response to be regime-dependent (leading per decade inside the box, nearly flat at routine defects), replaces an analytic resistance derate with scripted parasitic sweeps, checks the doping by Mott–Schottky self-consistency, and bounds the result by a model-derived radiative Voc that leaves the no-interface band without margin. Table 22 states the reporting standard as a concrete checklist. We recommend guardrailed reporting (interface status *with* dose level and defect level, defect floors *with* cross-sections, radiative coefficient *with* value, unit-system declaration, Rs basis, contact work functions, affinity choices) as a publication standard for SCAPS studies.

## Data availability

Every artefact named in this paper is archived in the public repository `https://github.com/touhidsiddiqueeraj-bit/cspbi3-cbts-guardrailed` at commit `6e991c1` (this draft). Definition files (`defs/`) carry the base cell with interfaces, the no-interface variant, and the work-function and cross-section variants; `outputs/` holds every sweep receipt and raw curve, including the per-run logs; `scripts/` regenerates each receipt from the archived definitions; `receipts/` and `manuscript/` hold the verification receipts and the typeset source. Table 23 is the itemised inventory. Reference PDFs under `sources/` are copyrighted and listed in REFERENCES.md instead. This exceeds the "available on reasonable request" convention: there is no request step, because the definition files reproduce every number in this paper from a clean checkout plus SCAPS-1D 3.3.10 under WINE.

## Author contributions

The authors designed the screening study, built the SCAPS definition files, executed the simulation campaign, performed the audit and sensitivity analyses, and wrote the manuscript.

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## Appendix A: two scripting traps and a verification rule

*Unit trap.* SCAPS definition files store SI units (metres, $m^{-3}$, $m^2$/Vs) but script `set` commands expect practical units ($\mu$m, $cm^{-3}$, $cm^2$/Vs)—verified by dose–response: `set layer2.NA 1e15` reproduces the file baseline (stored $10^{21}$ $m^{-3}$) exactly, while SI-valued inputs (e.g. $10^{22}$) inject a $10^6\times$ overdose and yield degenerate ghost cells (Voc 1.47–1.53 V, FF 91–92%, up to 27%). The same same-value check extends the map: `set layer2.radiative 0` and `set layer2.mun/mup 25/25` reproduce both headline receipts digit-identically, so radiative uses $cm^3$/s and mobilities $cm^2$/Vs in script units. Practical flags that catch them: Voc above the $E_g$-implied ceiling for the modelled gap, or FF above Green's ideal-diode value at the computed Voc.

*Structural trap.* `set interface<n>.IFdefect<k>.capture_cross_section` does not adjust $\sigma$ in place; it appears to restructure the interface defect block, and in our hands it neutralised the addressed interfaces entirely (one interface: 23.8565%; both interfaces: 25.1494%, essentially the no-interface value). The definition-file route (rewriting `sigma_nleft/.../pright`) reproduced the $N \times 10$ result to four decimals, which the set-key route could not.

*Rule.* For any parameter that changes a defect population, verify the applied value with a `get` of the same field in the same script and record the returned number in the run receipt; treat a result that moves efficiency in the wrong direction as a failed application, not a physical finding.

## Appendix B: run inventory

All simulation batches are inventoried in Tables 23 and 24.

**Table 23:** Simulation batches archived in the repository (run counts; solo reruns noted where they occurred).

| Batch | Configurations | Receipt |
|---|---|---|
| Baseline validation | 4 (one worker-crash retry) | `baseline_results.json` |
| Unit diagnostics | 6 + 4 retries | `diag_units.json` |
| Round A grid | 96 (3 worker-crash misses) | `roundA.jsonl` |
| Round B refine | 36 (3 worker-crash misses) | `roundB.jsonl` |
| Characterisation + audit + T | 17 + 6 re-runs | char.jsonl |
| Set-command verification | 5 | `s0_verify.json` |
| Interface dose + Rs/Rsh + WF | 14 | `s1_dose.json` |
| $N_t \times \sigma$ joint | 3 | `s4_sig.json` + `s4b_sig.json` |
| Morris screen | 56 (0 errors) | `s5_morris.jsonl` (precursor; range breaches floor) |
| Morris rerun, guardrail-clean | 56 (0 errors) | `s25_morris.jsonl` + `s25_analysis.json` |
| $\sigma$ verify + $\chi_{\mathrm{CBTS}}$ + Nt1e15 grid + realistic + noIF-JV | 16 (1 worker retry) | `s6_rev2.json` + `noIF_JV.json` |
| Affinity adoption ($\chi_{\mathrm{TiO2}}$) + adopted-base anchors | 12 | `s7_r3.json` + `s8_champ38.json` |
| Morris bootstrap intervals | 5000 resamples | `s5_morris_ci.json` |
| Routine-defect $\sigma \times 1$ baseline | 1 | `s9_nt15base.json` |
| Attainable gap + $E_t$ sweep | 6 | `s10_r4.json` |
| Fine-IV confirmatory (surveyed base) | 1 | `s12_fine.json` |
| Joint affinities + ITO/Ni | 52 (3 solo reruns merged) | `s11_joint_chi.json` |
| Box optimiser (LHS + pattern) | 104 (9 worker-file misses) | `s13_opt.json` |
| Paired Morris bootstrap + $\tau/S$ | analysis only | `s13_analysis.json` |
| Routine Rs + adopted Rs/Rsh + QE | 3 | `s14.json` + `adopted_qe_raw.txt` |
| Affinity-peak extension + new-base T/Eg/Rs | 26 + 6 (4 solo reruns merged) | `s16.json` + `s17.json` |

**Table 24:** Simulation batches archived (continued).

| Batch | Configurations | Receipt |
|---|---|---|
| Sobol-on-surrogate | RF, $N = 4096$ Saltelli | `sobol_surrogate.json` |
| Morris summary + CI + $\tau/S$ | derived, 5000 resamples | `s5_morris_summary.json` + `s5_morris_ci.json` |
| $N_t$ 1e12 step retry | 1 | `retry_nt.json` |
| Analytic: DB Voc, QE integral, SQ Jsc | closed form | `s18_analytic.json` |
| Probes + set-key verification + analysis | 4 + 5 + derived | `probe_qecv.json`, `s10_sigverify.json`, `analysis.json` |
| Thickness $\times$ $N_t$ factorial | 8 (2 solo reruns, deduped) | `s21_thNt.json` |
| Joint-optimum interface dose check | 2 | `s22_jointIF.json` |
| Stacked realistic variants + joint char + baseline JV | 5 + 3 + 1 | `s24_stack.json` + `baseline_JV.json` |
| Biased band diagrams | 2 | `s24_stack.json` (JEB_v115, JEB_v130) |
| Inert spectrum keys (documented failure) | 4 | `s24_stack.json` (J_NREL + spectrum tests; default IV returned digit-identically) |
| QE illumination-mode test | 2 | `s24_stack.json` (JQE_mode_test note; curves identical) |
| Contact artifact + DOS + Cr bracket + IV-step | 12 | `s26_diag.json` + 2 archived def variants |
| Cr $\times$ affinity map + extensions + routine subgrid | 90 (0 errors) | `s30_crchi.jsonl` |
| Screen self-consistency + source DOS + spectrum tests | 11 | `s31_small.json` |
| Calibration ladder | 7 (1 retry) | `s28_calib.json` |
| Prior-predictive LHS UQ + Cr0 sensitivity | 96 + 8 (1 divergence) | `s29_lhs.jsonl` + `s29_analysis.json` |
| Prior-predictive LHS + DOS dimension | 104 (0 errors) | `s32_lhs_dos.jsonl` + `s32_analysis.json` |
| Doping–affinity interaction | 4 (0 errors) | `s33_nachi.json` |
| Automated screen gates over all receipts | 1024 checked, 2 flagged | `s34_gates.json` |
| Radiative + mobility pricing (1 worker-crash retry) | 11 | `s23_krad_mu.json` |
| Raw per-run logs (jsonl) | – | `s13_opt.jsonl`, `s16.jsonl` |